Question Bank

Course Name: Statistics For Data Analytics-IM241AT

Sr No	Question with Solution & Scheme	Unit No	Topic Id	Topic Name	Difficulty Level	Marks	BT
1 1.1	**Question:** 40th percentile in a given set of 140 data values occurs at position _____ **Solution and Scheme:** 40th percentile in 140 data values Position = (40/100) × 140 = 56th Answer: 56th position	1	1	Data types and Charts	Moderate	1.00	3
2 1.2	**Question:** The median in the above data lies in the class _____ Class: 21"-30" | 31"-40" | 41"-50" | 51"-60" | 61"-70" No of occurrences: 12 | 16 | 22 | 25 | 18 **Solution and Scheme:** Cumulative frequencies: 12, 28, 50, 75, 93 → Median is at 47th position Answer: 41"-50"	1	1	Data types and Charts	Moderate	1.00	2
3 1.3	**Question:** Data on length of 93 wooden planks used in a carpentry shop is given below Length Classes in inches: 21-30 | 31-40 | 41-50 | 51-60 | 61-70 No of planks: 12 | 16 | 22 | 25 | 18 The mode in the above data lies in the class _____ **Solution and Scheme:** Since the highest frequency is 25, the mode lies in the class: 51–60 Answer: 51–60	1	1	Data types and Charts	Moderate	1.00	3
4 1.4	**Question:** For the given data set identify the median, and the first quartile, 62 , 65 , 68 , 70 , 72 , 74 , 76 , 78 , 80 , 82 , 96 , 101 **Solution and Scheme:** Median (Q2) = 75 First Quartile (Q1) = 69	1	1	Data types and Charts	Moderate	2.00	3
5 1.5	**Question:** For the given data set identify the third quartile, and the interquartile range (IQR). 62 , 65 , 68 , 70 , 72 , 74 , 76 , 78 , 80 , 82 , 96 , 101 **Solution and Scheme:**	1	1	Data types and Charts	Moderate	2.00	3

	Third Quartile (Q3) = 81 Interquartile Range (IQR) = 12						
6 1.6	**Question:** For the ordered data set: 38, 39, 41, 45, 46, 47, 48, 48, 51 what is the inter quartile range? **Solution and Scheme:** IQR=Q3−Q1=48−40=8	1	1	Data types and Charts	Moderate	2.00	3
7 1.7	**Question:** What is the median value for following ungrouped data: 86, 52,49,43,35,31,30,11 **Solution and Scheme:** Median = 39 : Median =(35 +43)/2	1	1	Data types and Charts	Easy	2.00	3
8 1.8	**Question:** What inference can be drawn from a data set with different values of mean, median and mode? **Solution and Scheme:** **Difference** among mean, median, and mode = **skewed distribution**. The **direction** of skewness can be identified by the order: • Mode < Median < Mean → **Right-skewed** • Mean < Median < Mode → **Left-skewed**	1	1	Data types and Charts	Moderate	2.00	2
9 1.9	**Question:** In what way is a stem and leaf diagram better than a histogram? **Solution and Scheme:** A stem-and-leaf diagram is better than a histogram because it retains the original data values while also showing the distribution and shape of the data. It is especially useful for small datasets and makes it easy to identify the median and mode quickly.	1	1	Data types and Charts	Easy	2.00	2
10 2.1	**Question:** A lot of 100 semiconductor chips contains 25 that are defective. Two are selected randomly, without replacement, from the lot. What is the probability that both are defective? _____ **Solution and Scheme:** (25/100 *24/99) =6/99 =2/33	1	2	Concepts of Probability	Moderate	2.00	3
11 2.2	**Question:** X = the no. coming up when a 6-faced die is thrown. Y = no. of heads when a coin is tossed. The possible values of X and Y are _____ **Solution and Scheme:** 0 to 6	1	2	Concepts of Probability	Easy	2.00	3
12 2.3	**Question:** A = {1, 2, 3, 4}; B = {2, 4, 6, 8}; C = {6, 7, 8, 9}; D = {4, 5, 6, 7}. If the sample space S = A U B U C U D, and each element of S is equally likely to happen, then P(A´∩ B) =__	1	2	Concepts of Probability	Moderate	2.00	2

	Solution and Scheme: S=A∪B∪C∪D={1,2,3,4,5,6,7,8,9}, total 9 A′∩B={6,8} → P = 2/9.						
13 2.4	**Question:** What is the sample space for the no. of calls repeatedly placed to a busy phone line until a connect is achieved? _____ **Solution and Scheme:** S={1,2,3,……..}	1	2	Concepts of Probability	Easy	1.00	3
14 2.5	**Question:** A part selected for testing is equally likely to have been produced on any one of six cutting tools. The probability that the part is not from tool 4 = _____ **Solution and Scheme:** P(Not Tool 4)=1−1/6=5/6	1	2	Concepts of Probability	Moderate	1.00	3
15 2.6	**Question:** Cooking oil is produced in two main varieties: mono and polyunsaturated. Two common sources of cooking oil are corn and canola. The following table shows the number of bottles of these oils at a supermarket: Type of unsaturation / Type of oil: Canola / Corn Mono: 7 / 13 Poly: 93 / 77 If a bottle of oil is selected at random, what is the probability that it belongs to the polyunsaturated category? **Solution and Scheme:** Total =190 Prob =170/190 =17/19 = 0.8947	1	2	Concepts of Probability	Moderate	2.00	4
16 2.7	**Question:** A = {1, 2, 3, 4}; B = {2, 4, 6, 8}; C = {6, 7, 8, 9}; D = {4, 5, 6, 7}. If the sample space S = A U B U C U D, and each element of S is equally likely to happen, then P(A´ | D) = _____ **Solution and Scheme:** S=A∪B∪C∪D={1,2,3,4,5,6,7,8,9} P(A′|D)=3/4	1	2	Concepts of Probability	Moderate	2.00	2
17 2.8	**Question:** In a throw of a fair 6-sided die, given that a prime no. comes up, what is the probability of a number at most 3 occuring? **Solution and Scheme:** Among the primes {2,3,5}, the numbers that are **at most 3** are:{2,3} Prob = 2/3	1	2	Concepts of Probability	Moderate	2.00	3

18 3.1	**Question:** A table with all possible value of a random variable and its corresponding probabilities is called _____ **Solution and Scheme:** Probability distributions	2	3	Concepts of Random Variables	Easy	1.00	2
19 2.9	**Question:** For a partitioning A, B & C of a set S and an event Y in S, what is P (A | Y), using Bayes theorem? **Solution and Scheme:** $P(A\|Y)=P(Y)P(Y\|A)\cdot P(A)$	1	2	Concepts of Probability	Moderate	1.00	2
20 2.10	**Question:** The probability distribution interms of the sample space and associated probabilities are as given below. Is this a valid distribution? Why or why not? **x** 1 2 3 4 **P(x)** 4/7 3/7 2/7 1/7 **Solution and Scheme:** No, this is not a valid probability distribution because the sum of the probabilities is greater than 1.	1	2	Concepts of Probability	Moderate	2.00	4
21 2.11	**Question:** In a throw of a fair 6-sided die, given that a non-prime number comes up, what is the probability of that number being odd? _____ **Solution and Scheme:** Non-prime numbers: 1, 4, 6 Odd among non-primes: 1 So, P(odd|non-prime)=1/3	1	2	Concepts of Probability	Moderate	2.00	3
22 1.10	**Question:** For the data set: 45, 41, 48, 39, 52, 38, 46, 51, 47, 48 what is the interquartile range? **Solution and Scheme:** IQR = Q3 – Q1 = 48 – 41 = 7	1	1	Data types and Charts	Moderate	2.00	3
23 1.11	**Question:** The weights (in pounds) of 30 newborn babies are listed below. Find Q1. 5.5 5.7 5.8 6.0 6.1 6.1 6.3 6.4 6.5 6.6 6.7 6.7 6.7 6.9 7.0 7.0 7.0 7.1 7.2 7.2 7.4 7.5 7.7 7.7 7.8 8.0 8.1 8.1 8.3 8.7 **Solution and Scheme:** Q1 = 31/4 = 7.75 position value = 6.3 + 0.075 = **6.375**	1	1	Data types and Charts	Moderate	2.00	3
24	**Question:**	2	3	Concepts of Random	Easy	2.00	2

	If f(x) is a probability density function of a continuous random variable, then =? **Solution and Scheme:** This is because the total probability over the entire range of a continuous random variable must be **1**.			Variables			
25 3.3	**Question:** The probability is p = 0.80 that a patient with a certain disease will be successfully treated with a new medical treatment. Suppose that the treatment is used on 40 patients. What is the "expected value" of the number of patients who are successfully treated? **Solution and Scheme:** The expected number of successfully treated patients is 32.(40*0.8 – np-binomial distribution)	2	3	Concepts of Random Variables	Moderate	2.00	3
26 3.4	**Question:** The sum of all probabilities in a probability mass function is equal to _____ **Solution and Scheme:** 1	2	3	Concepts of Random Variables	Easy	1.00	2
27 3.5	**Question:** The _____ function gives the probability that a discrete random variable is equal to a specific value. **Solution and Scheme:** Probability mass	2	3	Concepts of Random Variables	Easy	1.00	2
28 3.6	**Question:** Differentiate between discrete and continuous probability distributions? **Solution and Scheme:** Discrete distributions are defined for countable outcomes (e.g., binomial), whereas continuous distributions are for uncountable outcomes (e.g., normal distribution).	2	3	Concepts of Random Variables	Moderate	2.00	4
29 3.7	**Question:** For a continuous random variable, the probability that it takes an exact value is _____. **Solution and Scheme:** zero	2	3	Concepts of Random Variables	Easy	1.00	2
30 3.8	**Question:** The area under the probability density function (PDF) of a continuous random variable over its entire range is _____. **Solution and Scheme:** 1	2	3	Concepts of Random Variables	Easy	1.00	2
31 3.9	**Question:** A function f(x) is a valid probability mass function if ∑f(x)= _____ and f(x) ≥ _____ for all x.	2	3	Concepts of Random Variables	Easy	2.00	2

	Solution and Scheme: 1,0						
32 3.10	**Question:** Give an example of a situation that leads to a discrete probability distribution. **Solution and Scheme:** Rolling a die: The outcomes (1 to 6) are discrete and have defined probabilities.	2	3	Concepts of Random Variables	Moderate	2.00	2
33 4.1	**Question:** The arrivals of cyclists at the summit can be modelled with a Poisson process having intensity λ = 8 (cyclists per hour) Given that no cyclist arrived in the last half an hour, compute the probability that in the next half an hour 3 cyclists arrive. **Solution and Scheme:** (e^-4)*(4^3)/3! = 0.1953	2	4	Discrete Probability distribution	Easy	2.00	3
34 4.2	**Question:** The binomial distribution models the number of _____ in a fixed number of independent trials **Solution and Scheme:** successes	2	4	Discrete Probability distribution	Easy	1.00	2
35 4.3	**Question:** In a geometric distribution, the random variable represents the number of trials until the In a geometric distribution, the random variable represents the number of trials until the first _____.In a geometric distribution, the random variable represents the number of trials until the _____ **Solution and Scheme:** first success	2	4	Discrete Probability distribution	Easy	1.00	2
36 4.4	**Question:** If the number of calls received per hour at a call center is Poisson distributed with a mean of 4, what is the probability that exactly 2 calls are received in an hour? **Solution and Scheme:** P(X=2) ≈0.1465	2	4	Discrete Probability distribution	Moderate	2.00	3
37 4.5	**Question:** A Poisson process has a rate of 3 accidents per day. What is the probability of having **no** accidents on a given day? **Solution and Scheme:** P(X=0) ≈0.0498	2	4	Discrete Probability distribution	Moderate	2.00	3
38 4.6	**Question:** If the probability of success is 0.2, what is the probability that the first success occurs on the 4th trial?	2	4	Discrete Probability distribution	Moderate	2.00	3

	Solution and Scheme: Geometric P(X=4) = 0.1024						
39 4.7	**Question:** In the negative binomial distribution, the random variable counts the number of trials needed to achieve _____ successess **Solution and Scheme:** R or k	2	4	Discrete Probability distribution	Moderate	1.00	2
40 4.8	**Question:** A negative binomial distribution counts the number of trials needed to get 3 successes with p=0.4. What is the probability that the 3rd success occurs on the 5th trial? **Solution and Scheme:** 0.1382	2	4	Discrete Probability distribution	Moderate	2.00	2
41 5.1	**Question:** What is the shape of the graph of a continuous uniform distribution? **Solution and Scheme:** A horizontal straight line (rectangular).	3	5	Continuous Probability distribution	Moderate	1.00	2
42 5.2	**Question:** The area under the curve of a normal distribution represents the total probability and is always equal to ____. **Solution and Scheme:** 1	3	5	Continuous Probability distribution	Easy	1.00	2
43 5.3	**Question:** Enumerate any two real-life situations where the continuous uniform distribution can be applied. **Solution and Scheme:** • Time taken for a bus to arrive (if evenly distributed between 0–15 mins). • Random number generator between 0 and 1.	3	5	Continuous Probability distribution	Moderate	2.00	3
44 5.4	**Question:** What is meant by 'standard normal distribution'? **Solution and Scheme:** A normal distribution with mean μ=0 and standard deviation σ=1.	3	5	Continuous Probability distribution	Moderate	2.00	2
45 5.5	**Question:** What is the memoryless property of exponential distribution? **Solution and Scheme:**	3	5	Continuous Probability distribution	Hard	2.00	2

	The probability of an event occurring in the future is independent of the past. That is, P(X> s+t | X>s) =P(X>t).						
46 5.6	**Question:** In a customer service center, the time between arrivals follows an exponential distribution with mean 10 minutes. What is the probability that the next customer arrives **after** 15 minutes? **Solution and Scheme:** 0.2231	3	5	Continuous Probability distribution	Moderate	2.00	3
47 5.7	**Question:** A fair coin is tossed 100 times. Use normal approximation to find the probability of getting more than 60 heads. **Solution and Scheme:** Mean = 50, SD = 5 P(X>60) = 0.0179	3	5	Continuous Probability distribution	Moderate	2.00	3
48 5.8	**Question:** For a normal distribution with μ=70, σ=8, find P(62<X<78) **Solution and Scheme:** P(−1<Z<1)≈0.6826	3	5	Continuous Probability distribution	Moderate	2.00	2
49 5.9	**Question:** A random variable X is uniformly distributed over the interval [2,6]. Find the probability that X lies between 3 and 5. **Solution and Scheme:** P(3≤X≤5)= 5−3/6−2=2/4=0.5	3	5	Continuous Probability distribution	Moderate	2.00	2
50 5.10	**Question:** The mean for the exponential distribution equals the mean for the Poisson distribution only when the former distribution has a mean equal to **Solution and Scheme:** 1	3	5	Continuous Probability distribution	Easy	1.00	2
51 6.11	**Question:** The Central Limit Theorem states that, for large sample sizes, the sampling distribution of the sample mean is approximately _____, regardless of the population distribution. **Solution and Scheme:** Normal	3	6	Sampling Distribution	Easy	1.00	2
52 6.2	**Question:**	3	6	Sampling Distribution	Moderate	1.00	2

	As the sample size increases, the standard error of the mean will _____. **Solution and Scheme:** Decrease						
53 6.3	**Question:** The Central Limit Theorem is especially useful when the population distribution is _____. **Solution and Scheme:** Unknown, or non-normal	3	6	Sampling Distribution	Moderate	1.00	2
54 6.4	**Question:** What is the difference between population standard deviation and standard error? **Solution and Scheme:** Population standard deviation measures variability in the entire population, while standard error measures the variability of the sample mean across samples and is smaller than the population standard deviation	3	6	Sampling Distribution	Easy	2.00	2
55 6.5	**Question:** The mean weight of apples in an orchard is 150 grams with a standard deviation of 30 grams. A sample of 36 apples is taken. What is the probability that the sample mean is between 145 and 155 grams? **Solution and Scheme:** 68%	3	6	Sampling Distribution	Moderate	2.00	3
56 6.6	**Question:** Why is the sampling distribution important in statistics? **Solution and Scheme:** It allows us to make inferences about population parameters using sample statistics and compute probabilities.	3	6	Sampling Distribution	Moderate	2.00	2
57 6.7	**Question:** The standard deviation of a sampling distribution is also known as the _____. **Solution and Scheme:** Standard error	3	6	Sampling Distribution	Moderate	1.00	2
58 6.8	**Question:** If a population has a mean μ and standard deviation σ, the standard error of the mean is given by σ/sqrt{n}, where n is the _____. **Solution and Scheme:** **Answer:** sample size	3	6	Sampling Distribution	Easy	1.00	2

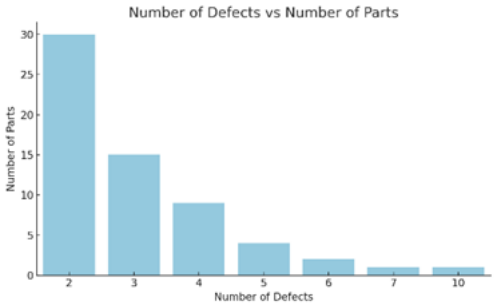
59 7.1	**Question:** In a regression analysis, the sum of square SXY = 385, and the relation is y = 2 + 5*x. The regression sum of squares is ___. **Solution and Scheme:** 1925	4	7	Simple linear Regression and Correlation	Hard	2.00	3
60 7.2	**Question:** The linear regression relation y = 2 + 3*x1 + 4*x2 has ___ independent variable(s). (x1 and x2 are variables with subscripts 1 and 2) **Solution and Scheme:** 2	4	7	Simple linear Regression and Correlation	Easy	1.00	2
61 7.3	**Question:** The residual in a linear regression model $\hat{y}_i = a + bx_i$ corresponding to an observed value y_i is _____ **Solution and Scheme:** $E = y_i - \hat{y}_i$	4	7	Simple linear Regression and Correlation	Moderate	1.00	2
62 7.4	**Question:** From Minitab output: Coefficients: Intercept = 2.5, Slope = 1.8, If X = 6, what is the predicted value of Y? **Solution and Scheme:** $Y = 2.5 + 1.8 \times 6 = 13.3$	4	7	Simple linear Regression and Correlation	Moderate	2.00	2
63 7.5	**Question:** Given the data: X: 1, 2, 3, 4 Y: 2, 4, 5, 4 Find the least squares estimates of β_0 and β_1. **Solution and Scheme:** $\beta_0 = 0.8,\ \beta_1 = 1.75$	4	7	Simple linear Regression and Correlation	Moderate	2.00	3
64 7.6	**Question:** The method of estimating the coefficients β_0 and β_1 by minimizing the sum of squared errors is called the _____ method. **Solution and Scheme:** Least squares	4	7	Simple linear Regression and Correlation	Moderate	1.00	2

65 7.7	**Question:** In simple linear regression, the coefficient of determination R^2 tells us the proportion of variation in _____ explained by the model. **Solution and Scheme:** The dependent variable (Y)	4	7	Simple linear Regression and Correlation	Moderate	1.00	2
66 7.8	**Question:** When regression is used to predict far outside the range of observed data, it is called _____. **Solution and Scheme:** Extrapolation	4	7	Simple linear Regression and Correlation	Moderate	1.00	2
67 7.9	**Question:** One common abuse of regression analysis is using it with _____ variables without checking for linearity. **Solution and Scheme:** categorical / non-quantitative	4	7	Simple linear Regression and Correlation	Hard	1.00	2
68 7.10	**Question:** State two assumptions of the simple linear regression model. **Solution and Scheme:** Linearity between X and Y Errors are normally distributed with constant variance (homoscedasticity)	4	7	Simple linear Regression and Correlation	Moderate	2.00	2
69 7.11	**Question:** What does a correlation coefficient of -0.85 indicate? **Solution and Scheme:** A strong negative linear relationship between the variables.	4	7	Simple linear Regression and Correlation	Moderate	2.00	2
70 7.12	**Question:** List one output from Minitab used to check the significance of a regression model. **Solution and Scheme:** P value for the slope coefficient (β_1)	4	7	Simple linear Regression and Correlation	Easy	2.00	2
71 7.13	**Question:** The strength of regression relation is given by ___ **Solution and Scheme:** Coefficient of Determination (R^2)	4	7	Simple linear Regression and Correlation	Easy	1.00	2

72 7.14	**Question:** In a regression analysis, the sum of squares S_{XY} = 385, S_{YY} = 2308.3, and the relation is y = 2 + 5*x. R-square = ____. **Solution and Scheme:** R-square = 0.8343	4	7	Simple linear Regression and Correlation	Moderate	2.00	3
73 7.15	**Question:** S_{XX} = 53, S_{XY} = 48, and S_{YY} = 84 in a regression analysis. Then SS_{Error} is given by _____. **Solution and Scheme:** SSE=84−48^2 /53 =40.51	4	7	Simple linear Regression and Correlation	Moderate	2.00	3
74 7.16	**Question:** When a set of points are used to fit a 2nd degree curve, _____ regression is said to be performed. **Solution and Scheme:** Quadratic regression	4	7	Simple linear Regression and Correlation	Moderate	1.00	2
75 7.17	**Question:** In linear regression the error term is distributed as _____ distribution (include its parameters also). **Solution and Scheme:** Normal distribution with mean 0 and constant variance σ^2.	4	7	Simple linear Regression and Correlation	Moderate	2.00	2
76 7.18	**Question:** When can you be confident that the regression relation is a proper one and can be used for prediction? **Solution and Scheme:** When the coefficient of determination (R^2) is high • The p-value for the slope is statistically significant (p < α, typically 0.05) • The residuals show no pattern (randomly distributed) and are normally distributed with constant variance • Assumptions of linear regression are satisfied (linearity, independence, homoscedasticity, normality)	4	7	Simple linear Regression and Correlation	Moderate	2.00	2
77 8.1	**Question:** The null hypothesis is rejected when the p-value is _____ the level of significance. **Solution and Scheme:** Less than <	5	8	Statistical Inference for a single sample, Interval Estimation	Easy	1.00	2

78 8.2	**Question:** Differentiate between the Type I and Type II errors in hypothesis testing. **Solution and Scheme:** Type I error: Rejecting a true null hypothesis. Type II error: Failing to reject a false null hypothesis.	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	2.00	2
79 8.3	**Question:** A sample mean is 52, population standard deviation is 6, and sample size is 36. Construct a 95% confidence interval for the population mean. **Solution and Scheme:** $CI=\bar{x}\pm Z\alpha/2\times\sigma/\sqrt{n}$ $=52\pm1.96\times6/\sqrt{36}=52\pm1.96\times1=(50.04,53.96)$	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	2.00	3
80 8.4	**Question:** If the confidence level is increased from 95% to 99%, the width of the confidence interval will be_____. **Solution and Scheme:** wider	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	1.00	2
81 8.5	**Question:** A machine fills juice bottles. The fill amount follows N(500, 16^2). A sample of 25 bottles had a mean of 505 ml. Construct a 95% CI for the population mean. **Solution and Scheme:** $\mu\in\bar{x}\pm Z\alpha/2*\sigma/(\sqrt{n})$ $=505\pm1.96*16/5 = 505\pm1.96*3.2 =(498.73,511.27)$	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	2.00	3
82 8.6	**Question:** A sample of 10 students has a mean score of 75 and a sample standard deviation of 5. Construct a 95% CI for the mean. **Solution and Scheme:** $CI=75\pm2.262*5/\sqrt{10} =75\pm3.58=(71.42, 78.58)$	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	2.00	3
83 8.7	**Question:** The test statistic used to test the population variance follows the _____ distribution. **Solution and Scheme:** Chisquare	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	1.00	2
84 8.7	**Question:**	5	8	Statistical Inference for a single sample,	Moderate	2.00	3

	A sample of 15 items has variance 20. Test if the population variance is 16 at 5% significance level. **Solution and Scheme:** 17.5			Interval Estimation			
85 8.9	**Question:** The 90% percent point of a χ^2variable with 9 degrees of freedom is _____ **Solution and Scheme:** $\chi_{0.90}$, df=9 ≈14.684	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	2.00	3
86 8.10	**Question:** The test statistic in a test for population variance based on a given sample of size n is _____ (also state what the terms are). **Solution and Scheme:** χ2=(n−1)s^2/σ^2 Where: • n = sample size • s^2 = sample variance • σ^2 = population variance under null hypothesis • The test statistic follows a chi-square distribution with (n − 1) degrees of freedom	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	2.00	2
87 8.11	**Question:** Type I error is committed when a _____ hypothesis is _____. **Solution and Scheme:** True null, rejected	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	2.00	3
88 8.12	**Question:** The null and alternate hypothesis statements to test whether a new process has reduced the population mean or not are: _____. **Solution and Scheme:** Null hypothesis (H_0): $\mu = \mu_0$ Alternate hypothesis (H_1): $\mu < \mu_0$	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	2.00	3
89 1.1	**Question:** 62 parts were analyzed for the number of defects with the following distribution: No. of defects, i: 2, 3, 4, 5, 6, 7, 10 No. of parts with i defects: 30, 15, 9, 4, 2, 1, 1 Plot a bar chart of the no. of defects vs no. of corresponding parts. Compute the mean, median, mode and the standard deviation for the no. of defects in a part. **Solution and Scheme:**	1	1	Data types and Charts	Moderate	8.00	3

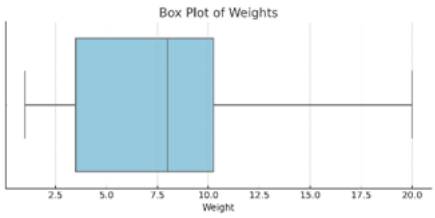


- Mean: 3.06
- Median: 3.0
- Mode: 2

Standard Deviation: 1.50

Plot -3 marks

Calculations - 5 marks

90 1.2	**Question:** 20 weight measurements in a process were recorded as: 2 8 11 10 8 4 2 2 9 8 4 11 12 2 4 4 20 1 10 15 (i) Draw a box plot for the data. (ii) What are the range and the IQR for the data? (iii) Are there any outliers? If yes, what are they? **Solution and Scheme:**  Range and IQR: • Minimum = 1 • Q1 (25th percentile) = 3.5 • Median (Q2) = 8.0 • Q3 (75th percentile) = 10.25 • Maximum = 20 • Range = 20 - 1 = 19 • IQR = Q3 - Q1 = 10.25 - 3.5 = 6.75 Outlier Bounds: • Lower bound = Q1 - 1.5×IQR = 3.5 - 10.125 = -6.625 • Upper bound = Q3 + 1.5×IQR = 10.25 + 10.125 = 20.375 Since all values lie between 1 and 20, there are no outliers in this dataset. Box plot - 7 marks including IQR and range outliers - 1 marks	1	1	Data types and Charts	Moderate	8.00	3
91 1.3	**Question:** The nine measurements that follow are furnace temperatures recorded on successive batches in a semiconductor manufacturing process (units are ^{0}F) 953, 950, 948, 955, 951, 949, 957, 954, and 955.	1	1	Data types and Charts	Moderate	8.00	3

a) Find the mode and median. How much could the largest temperature measurement increase without changing the median value?

b) Construct a box plot of the data, showing clearly the outlier bounds. Comment on the symmetry of the data.

Solution and Scheme:

Calculations:

- Mode and Median:
- Mode: 955 (appears twice)
- Median: 953.0

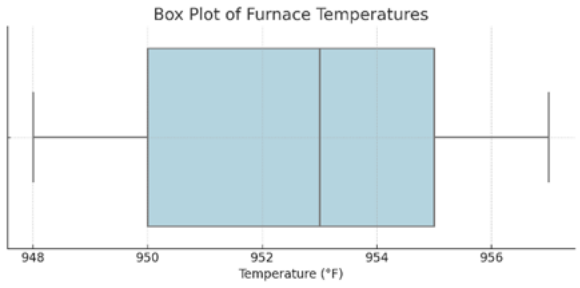
Since the median is the 5th value (953) in the sorted list and the largest value (957) is not affecting the position of the median, it can increase without changing the median — there is no upper limit on how much it can increase. --- 3 marks

b) Box Plot Observations:

- Q1 (1st Quartile): 949.0
- Q3 (3rd Quartile): 955.0
- Interquartile Range (IQR): 6.0
- Lower Bound for Outliers: 942.5
- Upper Bound for Outliers: 962.5
- Outliers: None — all values lie within the bounds.

Symmetry:

- The data is fairly symmetric with a slight right skew as the mean is slightly greater than the median.



---- 5 marks

92 1.4	**Question:**	1	1	Data types and Charts	Moderate	8.00	3

The no of defects when 25 parts were analyzed were recorded as: 15, 18, 20, 9, 11, 8, 16, 14, 13, 15, 17, 21, 7, 9, 11, 8, 18, 12, 6, 19, 15, 22, 19, 12, 10

a. Draw the box plot for the above data

b. Plot the stem and leaf diagram and comment on the data

Solution and Scheme:



STEM AND LEAF DIAGRAM

0 | 6 7 8 8 9 9

1 | 0 1 1 2 2 3 4 5 5 5 6 7 8 8 9 9

2 | 0 1 2

Box Plot Summary and Statistics:

- Q1 (25th percentile): 10.0
- Q3 (75th percentile): 18.0
- IQR (Interquartile Range): 8.0
- Lower Bound for Outliers: -2.0
- Upper Bound for Outliers: 30.0
- Outliers: None

Comment on Distribution:

- The data is fairly symmetrical, as the box plot shows a balanced whisker length on both sides.
- No outliers were detected.

The data ranges from 6 to 22, with the middle 50% of values (IQR) lying between 10 and 18

93 1.5	**Question:**	1	1	Data types and Charts	Moderate	8.00	3

The no of defects when 20 parts were analyzed were recorded as:

15, 18, 20, 29, 31, 8, 16, 14, 13, 25, 49, 11, 38, 18, 42, 9, 19, 15, 22, 19.

Draw the following plots and comment on the data using each plot.

(i) dot plot

(ii) stem and leaf diagram

(iii) box plot

(i

Solution and Scheme:

(i) Dot Plot

- Each dot represents one observation.
- It helps visualize how the data points are spread across the range.

0 | 8 9

1 | 1 3 4 5 5 6 8 8 9 9

2 | 0 2 5 9

3 | 1 8

4 | 2 9

- The *stem* (left of |) represents the tens place.
- The *leaf* (right of |) represents the units place.
- It shows a concentration in the 10–20 range.

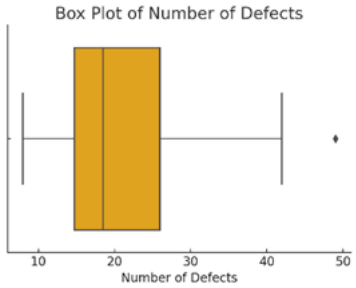
Box Plot Summary

- Q1 (25th percentile): 14.75
- Q3 (75th percentile): 26.00
- IQR (Q3 - Q1): 11.25
- Lower Bound: -2.125
- Upper Bound: 42.875
- Outliers: [49]

Comment on Distribution:

- The data is right-skewed due to the outlier 49.
- Most values lie between 10 and 30.

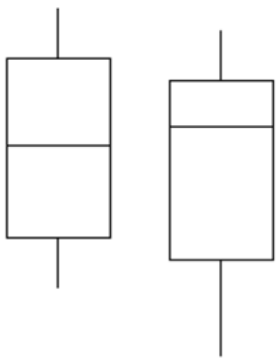
There is one outlier beyond the upper whisker (49).



Dot plot - 2marks

Stem and leaf - 3 marks

Box plot - 3marks

94 1.6	**Question:** Differentiate between nominal, ordinal, interval, and ratio data types with examples **Solution and Scheme:** • Nominal: Categories with no order (e.g., gender, blood type) • Ordinal: Ordered categories (e.g., satisfaction levels) • Interval: Numeric, no true zero (e.g., temperature in °C) • Ratio: Numeric, true zero (e.g., height, weight) Each 2 marks	1	1	Data types and Charts	Easy	8.00	2
95 1.7	**Question:** What are the key components of a radar chart? Where is it used? **Solution and Scheme:** • Axes for each variable • Radial grid lines • Polygon showing performance Used for comparing multi-attribute data (e.g., employee skills) --- 8 marks	1	1	Data types and Charts	Easy	8.00	2
96 1.8	**Question:** Describe any similarities or differences in the two distributions represented by the following boxplots. Assume the two boxplots have the same scale.  **Solution and Scheme:** **Feature** | **Left Boxplot** | **Right Boxplot** Median position | Roughly centered in the box | Closer to the top (Q3) of the box	1	1	Data types and Charts	Hard	8.00	2

Feature	Left Boxplot	Right Boxplot
IQR (Box height)	Symmetrical; equal distance from Q1–Q3	Same height, but the box is skewed upward
Skewness	Symmetric distribution	Negatively skewed (longer lower whisker)
Whisker Length	Whiskers of nearly equal length	Longer lower whisker than upper
Central Tendency	Median centered	Median closer to Q3

Any 4 differences - each difference - 2marks

97 1.9	**Question:** Suppose the excess length in a manufactured part in mm is recorded as below. 19 22 6 29 57 14 12 24 6 37 19 19 18 12 11 17 15 17 13 56 i. Compute the mean, median and mode for the data set. ii. Draw a histogram with 4 classes and iii. Draw a box plot, and interpret. **Solution and Scheme:** Mean:22.8 Median:=17.5 Mode=19 --- 2 marks Histogram (with 4 Classes) Range: • Min = 6, Max = 57 → Range = 51 • Class width = =13 Class Interval: Frequency 6 – 18: 10 19 – 31: 7 32 – 44: 1 45 – 57: 2 skewed distribution. --- 2marks Box Plot Using the ordered data: • Min = 6 • Q1 (25th percentile) = 12.5 • Median (Q2) = 17.5 • Q3 (75th percentile) = 23 • Max = 57 Outlier check (using IQR method): IQR=Q3−Q1=23−12.5=10.5 Lower bound: Q1−1.5×IQR=12.5−15.75=−3.25 Upper bound: Q3+1.5×IQR=23+15.75=38.75 Outliers: 56, 57 → 4 marks	1	1	Data types and Charts	Moderate	8.00	3
98	**Question:** Experiments were conducted to study the effectiveness of a braking system of a standard car. When the car was moving at a speed of 45 KMPH, brakes were applied and the stopping distances of car were measured, the stopping distances of car, in meters are given below.	1	1	Data types and Charts	Moderate	8.00	3

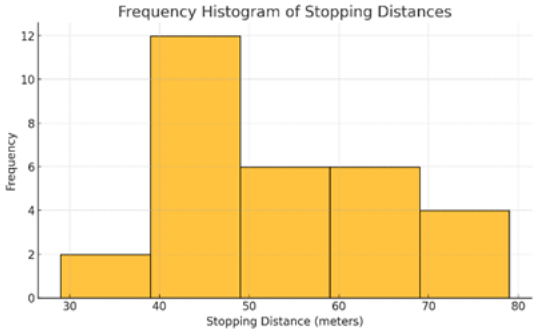
76	70	63	53	55	62	47	30	64	42
39	56	43	66	44	48	70	71	45	47
40	41	29	42	58	44	51	60	62	49

(i) Plot the frequency histogram.

(ii) Compute the mean, median and standard deviation.

(iii) Draw the stem and leaf diagram

Solution and Scheme:



-- 2marks

Mean: 52.23 meters

Median: 50.0 meters

Standard Deviation: 12.25 meters -- 3 marks

2 | 9

3 | 0 9

4 | 0 1 2 2 3 4 4 5 7 7 8 9

5 | 1 3 5 6 8

6 | 0 2 2 3 4 6

7 | 0 0 1 6

Each line shows the *tens* digit as the stem and the *units* digit(s) as the leaves.

The distribution is **slightly right-skewed**, given the presence of some higher values (like 76).

Most values fall within the **39–58** range, which is the central tendency.

No extreme outliers are immediately evident based on the histogram and stem-and-leaf diagram. ----3 marks

99	**Question:**	1	2	Concepts of Probability	Moderate	8.00	3

Customers evaluate products and either give them good reviews or don't. Also, the products are highly successful, moderately successful or poor according to sales performance. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and the rest have been poor products.

a) Depict the sample space with appropriate partitions in a sketch. What is the probability that a product attains a good review?

b) If a product attains a good review, what is the probability that it will be a highly successful product?

c) If a product does not attain a good review, what is the probability that it will be a highly successful product?

Solution and Scheme:

Part	Question	Answer
(a)	Probability product gets good review	0.615
(b)	Probability product is highly successful given good review	0.6179
(c)	Probability product is highly successful given **no** good review	0.0519

2 marks each

100

Question:

Consider the following random experiments and define the sample space for each of them. Also identify the type of sample space resulting from the experiment.

i. An observer conducting a traffic survey is interested in getting the frequency of traffic offence during peak hours.

ii. A chemical analyst recording the pH value of water using a PH meter.

iii. An automobile manufacturer is interested in counting the number of model variants created using three capacities (800 cc, 1000 cc, 1500 cc), two options (with AC, without AC), three colors (steel grey, Beige and peach) and three types fueling systems (petrol, Diesel, Hybrid)

iv. The time until a transaction service is requested of a computer

1 | 2 | Concepts of Probability | Moderate | 8.00 | 3

Solution and Scheme:

Expt No.	Description	Sample Space	Type
(i)	Frequency of traffic offences	{0,1,2,3,… }	Countably infinite (discrete)
(ii)	pH value recording	[0,14]⊂R[0, 14] \subset \	Uncountably infinite (continuous)
(iii)	Automobile variants	54 combinations of 4 attributes	Finite discrete
(iv)	Time until service request	(0,∞)⊂R	Uncountably infinite (continuous)

Each 2 marks

101

Question:

Pouches of a specialty chemical are tested for presence of impurities A or B, which makes the chemical unfit for use. The fitness for use is tested separately. Suppose 10% of the pouches have impurity A, 5% have impurity B, 1% have both impurities A & B, and other pouches have no impurities. Pouches are unfit for use with probabilities 1%, 2%, 6%, and 0.1% if they have impurity A, impurity B, both impurities A & B, and no impurities respectively. What is the probability that a randomly chosen pouch is unfit for use?

Solution and Scheme:

Let:

- A: Impurity A → P(A) = 0.10
- B: Impurity B → P(B) = 0.05
- A ∩ B = 0.01
- No impurity = 1 - (0.10 + 0.05 - 0.01) = 0.86 ----- 1 mark

Probabilities of unfitness:

- P(U|A only) = 0.01
- P(U|B only) = 0.02
- P(U|A ∩ B) = 0.06
- P(U|no impurity) = 0.001 ---------- 4 marks

P(U)=0.09*0.01+0.04*0.02+0.01*0.06+0.86*0.001=0.0009+0.0008+0.0006+0.00086 =0.00316 ----- 3 marks

1 | 2 | Concepts of Probability | Easy | 8.00 | 3

102

Question:

Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

		Shock resistance	
		high	low
Scratch	high	70	9
Resistance	low	16	5

1 | 2 | Concepts of Probability | Easy | 8.00 | 3

	Let *A* denote the event that a disk has high shock resistance, and let *B* denote the event that a disk has high scratch resistance. If a disk is selected at random, determine the following probabilities: (a) P(A) (b) P(B) (c) P(A') (d) P(A∩B) (e) P(A∪B) (f) P(A'∪B) (g) P(A/B) (h) P(B/A) **Solution and Scheme:** (a) P(A) = (70 + 16)/100 = **0.86** (b) P(B) = (70 + 9)/100 = **0.79** (c) P(A') = 1 - 0.86 = **0.14** (d) P(A ∩ B) = 70/100 = **0.70** (e) P(A ∪ B) = P(A) + P(B) - P(A ∩ B) = 0.86 + 0.79 - 0.70 = **0.95** (f) P(A' ∪ B) = 1 - P(A ∩ B') = 1 - (16/100) = **0.84** (g) P(A|B) = P(A ∩ B)/P(B) = 70/79 ≈ **0.886** (h) P(B|A) = 70/86 ≈ **0.814** **Each calculation - 1 mark**						
103	**Question:** In a certain factory, machines 1, 2, and 3 are all producing pins. Of their production, machines 1,2, and 3 produce 2%, 1%, and 3% defective pins respectively. Of the total production of pins, machine 1 produces 35%, machine 2 produces 25% and machine 3 produces 40%. If one pin is selected at random from the total pins what is the probability that it is defective **Solution and Scheme:** (a) Probability a pin is defective 0.0215 ---- 2 marks	1	2	Concepts of Probability	Moderate	2.00	3
104	**Question:** The rise time of a reactor is measured in minutes (and fractions of times). Let the sample space positive, real numbers. Define the events A and B as follows: $A=\{x \mid x<72.5\}$ and $B=\{x \mid x>52.5\}$ Describe each of the following events. i. A′ ii. B′ iii. A∩B iv. A∪B **Solution and Scheme:** i. $A' = \{x \mid x \geq 72.5\}$ ii. $B' = \{x \mid x \leq 52.5\}$ iii. $A \cap B = \{x \mid 52.5 < x < 72.5\}$ iv. $A \cup B = \{x \mid x > 0\}$	1	2	Concepts of Probability	Moderate	8.00	3
105	**Question:** Suppose it is known that 1% of the population suffers from a particular disease. A blood test has a 97% chance of a positive blood test (identifying the disease) for diseased individuals, but also has a 6% chance of falsely indicating that a healthy person has the disease. (i) What is the probability that a person will have a positive blood test? (ii) If your blood test is positive, what is the chance that you have the disease? **Solution and Scheme:**	1	2	Concepts of Probability	Moderate	8.00	4

$$P(\text{Positive test}) = P(\text{Positive}|D) \cdot P(D) + P(\text{Positive}|\text{No D}) \cdot P(\text{No D})$$
$$= (0.97 \times 0.01) + (0.06 \times 0.99)$$
$$= 0.0097 + 0.0594 = 0.0691$$

$$\boxed{P(\text{Positive test}) = 0.0691}$$

(ii) If your blood test is positive, what is the chance that you have the disease?

This is asking for the **posterior probability** using **Bayes' Theorem**:

$$P(D|\text{Positive}) = \frac{P(\text{Positive}|D) \cdot P(D)}{P(\text{Positive})}$$
$$= \frac{0.97 \times 0.01}{0.0691} = \frac{0.0097}{0.0691} \approx 0.1404$$

So,

$$\boxed{P(D|\text{Positive}) \approx 0.1404 \text{ or } 14.04\%}$$

106	**Question:** A smartphone manufacturing company inspects **100 mobile phone screens** for **display quality** and **touch sensitivity** before shipment. The inspection results are summarized below: Let: A denote event that a screen has high touch sensitivity and B event that a screen has high display quality. If a screen is selected at random, determine the following probabilities: (a) P(A) (b) P(B) (c) P(A') (d) P(A∩B) (e) P(A∪B) (f) P(A'∪B) (g) P(A/B) (h) P(B/A) **Solution and Scheme:** (a) P(A) = 0.82 (b) P(B) =0.80 (c) P(A') =0.18 (d) P(A∩B) =0.68 (e) P(A∪B) =0.94 (f) P(A'∪B) =0.86 (g) P(A/B) = 0.85 (h) P(B/A) =0.83 Each 1 mark	1	2	Concepts of Probability	Moderate	8.00	3

	Touch Sensitivity High	Touch Sensitivity Low	Total
Display Quality High	68	12	80
Display Quality Low	14	6	20
Total	82	18	100

107	**Question:** The probability density function of the weight of packages delivered by a post office is $f(x) = 70/(69x^2)$ for $1 < x < 70$ pounds. i) What is the probability a package weighs less than 10 pounds? ii) Determine the mean and variance of package weight. iii) If the shipping cost is $3 per pound, what is the average shipping cost of a package? **Solution and Scheme:** package weighs less than 10 pounds is 0.9130 (or 91.30%) --- 4 marks Mean ≈ 4.31 pounds Variance ≈ 51.42 --- 2 marks	2	3	Concepts of Random Variables	Moderate	8.00	3

	Average shipping cost = $12.93 ---2 marks						
108	**Question:** If a random variable X has its cumulative distribution as F(x) = 0, -∞ < x < 1, 0.4, 1 ≤ x < 3, 0.5, 3 ≤ x < 8, 1.0, 8 ≤ x < ∞ Determine the following: (i) P(X at most 3) (ii) E(X) (iii) P(X between 1 and 2, both inclusive) (iv) P(X at least 2) **Solution and Scheme:** (i) $P(X \leq 3)$ From CDF or summing PMF: $P(X \leq 3) = P(X = 1) + P(X = 3) = 0.4 + 0.1 = \boxed{0.5}$ (ii) $E(X)$ — **Expected Value** $E(X) = \sum x_i \cdot P(X = x_i) = 1(0.4) + 3(0.1) + 8(0.5) = 0.4 + 0.3 + 4.0 = \boxed{4.7}$ (iii) $P(1 \leq X \leq 2)$ Only $X = 1$ lies in this interval (since values are discrete): $P(1 \leq X \leq 2) = P(X = 1) = \boxed{0.4}$ (iv) $P(X \geq 2)$ That means $X = 3$ or $X = 8$: $P(X \geq 2) = P(X = 3) + P(X = 8) = 0.1 + 0.5 = \boxed{0.6}$ Each answer 2 marks	2	3	Concepts of Random Variables	Easy	8.00	3
109	**Question:** A discrete random variable XXX has the following probability mass function: **x** | **1** | **2** | **3** | **4** P(X=x) | 0.1 | 0.2 | 0.4 | 0.3 a)Find the cumulative distribution function (CDF) of X. b) Calculate E(X) and Var(X). **Solution and Scheme:** $F(1) = P(X \leq 1) = 0.1$ $F(2) = P(X \leq 2) = 0.1 + 0.2 = 0.3$ $F(3) = 0.3 + 0.4 = 0.7$ $F(4) = 0.7 + 0.3 = 1.0$ $F(x) = \begin{cases} 0 & x < 1 \\ 0.1 & 1 \leq x < 2 \\ 0.3 & 2 \leq x < 3 \\ 0.7 & 3 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$ ------ 5 marks E(X)=2.9, Var(X)=0.89 ----- 3 marks	2	3	Concepts of Random Variables	Moderate	8.00	3
110	**Question:**	2	3	Concepts of Random Variables	Moderate	8.00	3

Marketing estimates that a new instrument for the analysis of soil samples will be very successful, moderately successful, or unsuccessful, with probabilities 0.3, 0.6, and 0.1, respectively. The yearly revenue associated with a very successful, moderately successful, or unsuccessful product is $10 million, $5 million, and $1 million, respectively. Let the random variable X denote the yearly revenue of the product. Determine the probability mass function of X, and thus compute its mean and variance.

Solution and Scheme:

- Very Successful → Revenue = $10 million → Probability = 0.3
- Moderately Successful → Revenue = $5 million → Probability = 0.6
- Unsuccessful → Revenue = $1 million → Probability = 0.1

Define the Probability Mass Function (PMF) of X

$$X = \begin{cases} 10 & \text{with probability } 0.3 \\ 5 & \text{with probability } 0.6 \\ 1 & \text{with probability } 0.1 \end{cases}$$

So, the PMF is:

$$P(X = x) = \begin{cases} 0.3 & \text{if } x = 10 \\ 0.6 & \text{if } x = 5 \\ 0.1 & \text{if } x = 1 \\ 0 & \text{otherwise} \end{cases}$$

------- 5 marks

E(x) = 6.1 Million dollars

Variance = 45.1 --- 3 marks

111	**Question:** A random variable X is distributed as f(x) = x/k where Range of X = {1,2,3,4} i. What is the probability that X will take on a value of at most 2? ii. Plot the probability mass function of X. iii. Derive the cumulative distribution of X and plot the same. iv. Find out the mean and variance of X. **Solution and Scheme:** • $f(x) = \frac{x}{10}, x = 1, 2, 3, 4$ • $P(X \leq 2) = 0.3$ • CDF: {0.1, 0.3, 0.6, 1.0} • Mean = 3.0, Variance = 1 Each answer 2 marks	2	3	Concepts of Random Variables	Moderate	8.00	3
112	**Question:** A continuous random variable XXX has the probability density function $f(x) = \begin{cases} 2x, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$ **(i)** Find the probability that XXX lies between 0.25 and 0.75, i.e., P(0.25<X<0.75) **(ii)** Determine the mean E(X) and variance Var(X) **Solution and Scheme:**	2	3	Concepts of Random Variables	Moderate	8.00	3

a)

$$P(0.25 < X < 0.75) = \int_{0.25}^{0.75} 2x\,dx = [x^2]_{0.25}^{0.75} = (0.75)^2 - (0.25)^2 = 0.5625 - 0.0625 = \boxed{0.5}$$

b)

$$E(X) = \int_0^1 x \cdot 2x\,dx = \int_0^1 2x^2\,dx = \left[\frac{2x^3}{3}\right]_0^1 = \frac{2}{3}$$

$$E(X^2) = \int_0^1 x^2 \cdot 2x\,dx = \int_0^1 2x^3\,dx = \left[\frac{2x^4}{4}\right]_0^1 = \frac{1}{2}$$

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \boxed{\frac{1}{18}}$$

i) - 4 marks

ii) 4 marks

113	**Question:** A production facility records the number of machine breakdowns occurring in a week. Let the random variable *X*denote the number of breakdowns. The probability mass function (PMF) of *X*is given below: **(x)** | **0** | **1** | **2** | **3** | **4** | **5** (P(X=x)) | 0.10 | 0.20 | 0.30 | 0.15 | 0.15 | 0.10 Answer the following: (i) Determine the expected number of machine breakdowns, E(X) (ii) Find the probability that the number of machine breakdowns exceeds 3 **Solution and Scheme:** E(X) = 2.35 P(X>3) = P(X=4)+P(X=5) = 0.25 Each answer 4 marks	2	3	Concepts of Random Variables	Moderate	8.00	3
114	**Question:** Past experience indicates that the ticket office at the Rex Theatre receives a mean rate of thirty calls per hour. Assuming that the number of calls have the Poisson distribution, find the probability that a i) during a ten-minute coffee-break, no calls will be received ii) during a twenty-minute coffee-break, at least two calls will be received iii)During a half-an-hour lunch break, at most 3 calls will be received **Solution and Scheme:** **Probability that no calls will be received in a 10-minute coffee break** $\lambda = 30 \times \frac{10}{60} = 5$ $P(X = 0) = \frac{e^{-5} \cdot 5^0}{0!} = e^{-5} \approx 0.0067$ **Probability that at least 2 calls will be received in a 20-minute break** $\lambda = 30 \times \frac{20}{60} = 10$ $P(X \geq 2) = 1 - P(0) - P(1)$ $P(0) = \frac{e^{-10} \cdot 10^0}{0!} = e^{-10} \approx 4.54 \times 10^{-5}$ $P(1) = \frac{e^{-10} \cdot 10^1}{1!} = 10 \cdot e^{-10} \approx 4.54 \times 10^{-4}$ $P(X \geq 2) = 1 - (4.54 \times 10^{-5} + 4.54 \times 10^{-4}) = 1 - 5.0 \times 10^{-4} = \boxed{0.9995}$	2	4	Discrete Probability distribution	Hard	8.00	3

$$P(X \leq 3) = P(0) + P(1) + P(2) + P(3)$$

Using:

$$P(k) = \frac{e^{-15} \cdot 15^k}{k!}$$

- $P(0) = e^{-15} \cdot \frac{15^0}{0!} = e^{-15} \approx 3.06 \times 10^{-7}$
- $P(1) = e^{-15} \cdot \frac{15^1}{1!} = 15e^{-15} \approx 4.59 \times 10^{-6}$
- $P(2) = e^{-15} \cdot \frac{15^2}{2!} = 112.5e^{-15} \approx 3.44 \times 10^{-5}$
- $P(3) = e^{-15} \cdot \frac{15^3}{3!} = 562.5e^{-15} \approx 1.72 \times 10^{-4}$

$$P(X \leq 3) \approx 3.06 \times 10^{-7} + 4.59 \times 10^{-6} + 3.44 \times 10^{-5} + 1.72 \times 10^{-4} = \boxed{0.000214}$$

115	**Question:** The number of flaws in bolts of cloth in textile manufacturing is assumed to be Poisson distributed with a mean of 0.1 flaw per square meter. (i) What is the probability that there are two flaws in 1 square meter of cloth? (ii) What is the probability that there is one flaw in 10 square meters of cloth? (iii) What is the probability that there are no flaws in 20 square meters of cloth? **Solution and Scheme:** a) Let X denote the number of flaws in one square meter of cloth. Then, X is a Poisson random variable with $\lambda = 0.1$. $P(X = 2) = \frac{e^{-0.1}(0.1)^2}{2!} = 0.0045$ b) Let Y denote the number of flaws in 10 square meters of cloth. Then, Y is a Poisson random variable with $\lambda = 1$. $P(Y = 1) = \frac{e^{-1}1^1}{1!} = e^{-1} = 0.3679$ c) Let W denote the number of flaws in 20 square meters of cloth. Then, W is a Poisson random variable with $\lambda = 2$. $P(W = 0) = e^{-2} = 0.1353$ i) -- 2marks ii) 3marks iii) 3 marks	2	4	Discrete Probability distribution	Moderate	8.00	3
116	**Question:** Based on field observation it is found that the volume of traffic flow passing a point of highway segment is 720 vehicles per hour i) Calculate the probability that there is no vehicle passing this point in 5 seconds, assuming the arrival of vehicles are independent of each other and follows Poisson distribution. ii)Calculate the probability of exact 5 vehicles passing this point in 5 seconds. iii)Calculate the probability of less than 5 vehicles passing this point in 5 seconds. **Solution and Scheme:** $P(X = 0) = \frac{e^{-1} \cdot 1^0}{0!} = e^{-1} = 0.3679$ $P(X = 5) = \frac{e^{-1} \cdot 1^5}{5!} = \frac{e^{-1}}{120} \approx \frac{0.3679}{120} \approx 0.00307$ $P(X < 5) \approx 0.3679 + 0.3679 + 0.1839 + 0.0613 + 0.0153 = 0.9963$ i) 2marks ii) 3 marks iii) 3 marks	2	4	Discrete Probability distribution	Moderate	8.00	3
117	**Question:**	2	4	Discrete Probability distribution	Moderate	8.00	3

A manufacturer of LED bulbs finds that 8% of the bulbs produced are defective due to filament issues. A quality inspector randomly selects bulbs for inspection.

Answer the following:

(i) What is the probability that none of the defective bulbs is found in a sample of 12 bulbs?

(ii) What is the probability that at least one defective bulb is found in a sample of 25 bulbs?

(iii) What is the probability that exactly two defective bulbs are found in a sample of 15 bulbs?

Assume that the production process is stable and that bulb selections are independent.

Solution and Scheme:

(i) None defective in 12 bulbs	**0.368**
(ii) At least one defective in 25 bulbs	**0.876**
(iii) Exactly two defective in 15 bulbs	**0.228**

Each answer - 2, 3 and 3 marks

118

Question:

Suppose X has a poisson distribution with a mean of 4. Determine the following probability

i) P(X=0)

ii) P(X=8)

iii) P(X≤2)

iv) P(X=4)

Solution and Scheme:

a) $P(X=0)=\frac{e^{-4}4^{0}}{0!}=e^{-4}=0.0183$

b) $P(X\le 2)=P(X=0)+P(X=1)+P(X=2)$

$$=e^{-4}+\frac{e^{-4}4^{1}}{1!}+\frac{e^{-4}4^{2}}{2!}=0.2381$$

c) $P(X=4)=\frac{e^{-4}4^{4}}{4!}=0.1954$

d) $P(X=8)=\frac{e^{-4}4^{8}}{8!}=0.0298$

2 | 4 | Discrete Probability distribution | Moderate | 8.00 | 3

119

Question:

A factory producing washing machines sends 1,000 machines each month to one of their customers. The manufacturer's past records show that 5% of their machines are defective. The customer decides to pick a sample of 20 machines and subject them to tests.

i) What is the probability of the customer finding at most no defective in the sample?

ii) the customer tolerates at most 1 defective in the sample for accepting the consignment of 1,000 machines. What is the probability of the customer accepting the whole consignment?

iii) What is the mean number of defective machines in the sample?

Solution and Scheme:

$$P(X=0)=\binom{20}{0}(0.05)^{0}(0.95)^{20}=(0.95)^{20}$$

$$P(X=0)\approx 0.3585$$

$$P(X\le 1)=0.3585+0.3774=0.7359$$

$$\mu=n\cdot p=20\cdot 0.05=1$$

2 | 4 | Discrete Probability distribution | Moderate | 8.00 | 3

120	**Question:** A multiple-choice test contains 25 questions, each with four answers. Assume a student just guesses on each question. (i) What is the probability that the student answers at least 20 questions correctly? (ii) What is the probability the student answers at most 5 questions correctly? **Solution and Scheme:** Let X denote the number of questions answered correctly. Then, X is binomial with n = 25 and p = 0.25. $a) P(X \geq 20) = \binom{25}{20}0.25^{20}(0.75)^5 + \binom{25}{21}0.25^{21}(0.75)^4 + \binom{25}{22}0.25^{22}(0.75)^3 + \binom{25}{23}0.25^{23}(0.75)^2 + \binom{25}{24}0.25^{24}(0.75)^1 + \binom{25}{25}0.25^{25}(0.75)^0 \cong 0$ $b) P(X < 5) = \binom{25}{0}0.25^{0}(0.75)^{25} + \binom{25}{1}0.25^{1}(0.75)^{24} + \binom{25}{2}0.25^{2}(0.75)^{23} + \binom{25}{3}0.25^{3}(0.75)^{22} + \binom{25}{4}0.25^{4}(0.75)^{21} = 0.2137$	2	4	Discrete Probability distribution	Moderate	8.00	3
121	**Question:** Suppose that the log-ons to a computer network follow a Poisson process with an average of five counts per minute. i) What is the mean time between log-ons? ii) Determine the probability density and cumulative distribution functions for the time between log-ons? iii) Determine x such that the probability that at least one log-on occurs before time x minutes is 0.95. iv) What is the probability that more than 20 log-ons occur in 5 minutes? **Solution and Scheme:** Here, we have a Poisson process with parameter $\lambda = 5$ log-ons per minute. **a)** Let T denote the time until the next log-on. Then T is an exponential random variable and its mean is $E(T) = \frac{1}{5}$ **b)** The probability density function and cumulative distribution functions are $f(t) = 5e^{-5t},\ t \geq 0$, $F(t) = 1 - e^{-5t},\ t \geq 0$ **c)** $P(T \leq x) = 0.95 \Rightarrow 1 - e^{-5x} = 0.95 \Rightarrow \mathbf{x = -\frac{\ln(0.05)}{5}}$ **d)** Let X denote the number of log-ons in 20 minutes. Then X is Poisson with $\lambda = 25$. $P(X > 20) = 1 - P(X \leq 20) = 1 - \sum_{x=0}^{20} \frac{e^{-25}(25)^x}{x!} = \mathbf{0.815}$ or, since $\lambda = 25 > 5$, $P(X > 20) = 1 - P(X \leq 20) \cong 1 - P\left(Z \leq \frac{20.5 - 25}{\sqrt{25}}\right) = 1 - P(Z \leq -0.9) = P(Z \leq 0.9) = \mathbf{0.816}$ 2 marks each answer	2	4	Discrete Probability distribution	Moderate	8.00	3
122	**Question:** The random variable X has a binomial distribution with n = 10 and p = 0.01. Determine the following probabilities. i) P (X = 5) ii) P(X ≤ 2) iii) P(X ≥ 9) iv) P(3 ≤ X < 5)	2	4	Discrete Probability distribution	Moderate	8.00	2

	Solution and Scheme: $P(X=5)=\binom{10}{5}0.5^5(0.5)^5=0.2461$ $P(X\le 2)=\binom{10}{0}0.5^0 0.5^{10}+\binom{10}{1}0.5^1 0.5^9+\binom{10}{2}0.5^2 0.5^8$ $=0.5^{10}+10(0.5)^{10}+45(0.5)^{10}=0.0547$ $P(X\ge 9)=\binom{10}{9}0.5^9(0.5)^1+\binom{10}{10}0.5^{10}(0.5)^0=0.0107$ $P(3\le X<5)=\binom{10}{3}0.5^3 0.5^7+\binom{10}{4}0.5^4 0.5^6$ $=120(0.5)^{10}+210(0.5)^{10}=0.3223$ Each answer 2 marks						
123	**Question:** Assume that each of your calls to a popular radio station has a probability of 0.02 of connecting, that is, of not obtaining a busy signal. Assume that your calls are independent. (i) What is the probability that your first call that connects is your tenth call? (ii) What is the probability that it requires more than five calls for you to connect? (iii) What is the mean number of calls needed to connect? **Solution and Scheme:** Let X denote the number of calls needed to obtain a connection. Then, X is a geometric random variable with p = 0.02 a) $P(X=10)=(1-0.02)^9 0.02=0.98^9 0.02=0.0167$ b) $P(X>5)=1-P(X\le 4)=1-[P(X=1)+P(X=2)+P(X=3)+P(X=4)]$ $=1-[0.02+0.98(0.02)+0.98^2(0.02)+0.98^3(0.02)]$ $=1-0.0776=0.9224$ c) E(X) = 1/0.02 = 50 Part a- 3marks Part b - 3marks Partc- 2 marks	2	4	Discrete Probability distribution	Moderate	8.00	3
124	**Question:** Inter arrival times in minutes between successive customers in a hotel is exponentially distributed with a mean of 2 minutes. i) Compute the probability that inter arrival time between two customers is at most 2 minutes ii) What is the probability that at least 2 customers arrive in one minute? iii) What is the probability of no customers arriving in a span of 3 minutes? **Solution and Scheme:** P(T<=2) = 0.6321 -- 3 marks P(X>=2) =0.0901 --- 3 marks P(Y=0)=0.2231 -- 2marks	3	5	Continuous Probability distribution	Moderate	8.00	3
125	**Question:** Suppose that a large conference room at a certain company can be reserved for no more than 4 hours. Both long and short meetings occur quite often. In fact, it can be assumed that the length X of a meeting is equally likely in the interval [1.5, 4]. i) What is the probability density function? ii) What is the probability that any given meeting lasts at least 3 hours?	3	5	Continuous Probability distribution	Moderate	8.00	3

	iii) Find the mean and variance of length X of the meetings **Solution and Scheme:** $f(x) = \begin{cases} \frac{1}{4-1.5} = \frac{1}{2.5} = 0.4 & \text{for } 1.5 \le x \le 4 \\ 0 & \text{otherwise} \end{cases}$ $\boxed{f(x) = \begin{cases} 0.4 & \text{for } 1.5 \le x \le 4 \\ 0 & \text{otherwise} \end{cases}}$ $P(X \ge 3) = \int_3^4 f(x)\,dx = (4-3)\cdot 0.4 = 1 \cdot 0.4 = 0.4$ Mean: $\boxed{2.75 \text{ hours}}$ Variance: $\boxed{0.5208 \text{ hours}^2}$ Each answer - 2marks						
126	**Question:** A certain type of storage battery lasts, on average, 3.0 years with a standard deviation of 0.5 year. Assuming that battery life is normally distributed, find i) the probability that a given battery will last less than 2.3 years. ii) the probability that a battery lasts between 3.8 and 4.5 years. iii) What is the life period which 90% of the batteries exceed? **Solution and Scheme:** a. P(X<2.3)=P(Z<−1.4) = 0.0808 -- 2marks b. P(1.6<Z<3.0)=0.9987−0.9452=0.0535 --- 3 Marks c. x=3.0+(−1.2816)(0.5) x=2.3592≈2.36 years ----- 3 Marks	3	5	Continuous Probability distribution	Moderate	8.00	3
127	**Question:** Suppose the response time X at a certain on-line computer terminal (the elapsed time between the end of users inquiry and the beginning of the systems response to that inquiry) has an exponential distribution with expected response time equal to 5 sec. Compute: i. E(X) ii. The probability that the response time is at most 10 sec iii. The probability that response time is between 5 and 10 sec iv. Give one more instance of the exponential distribution frequently used. **Solution and Scheme:** i) E(X) = 5 seconds **ii)** $P(X \le 10) = 1 - 0.1353 = \boxed{0.8647}$ **iii) P(5≤X≤10)=0.8647−0.6321=0.2326** iv) Time between arrivals of customers at a service center (e.g., bank, ATM, or call center), assuming arrivals follow a Poisson process. Each answer 2 marks	3	5	Continuous Probability distribution	Moderate	8.00	3
128	**Question:** The diameter of the dot produced by a printer is normally distributed with a mean diameter of 0.002 inch and a standard deviation of 0.0004 inch.	3	5	Continuous Probability distribution	Moderate	8.00	3

	i) What is the probability that the diameter of a dot exceeds 0.0026 inch? ii) What is the probability that a diameter is between 0.0014 and 0.0028 inch? iii) What standard deviation of diameters is needed so that the probability in part (b) is 0.995? **Solution and Scheme:** a) $P(X > 0.0026) = P\left(Z > \frac{0.0026 - 0.002}{0.0004}\right)$ $= P(Z > 1.5)$ $= 1\text{-}P(Z < 1.5)$ $= 0.06681.$ b) $P(0.0014 < X < 0.0026) = P\left(\frac{0.0014 - 0.002}{0.0004} < Z < \frac{0.0026 - 0.002}{0.0004}\right)$ $= P(-1.5 < Z < 1.5)$ $= 0.86638.$ c) $P(0.0014 < X < 0.0026) = P\left(\frac{0.0014 - 0.002}{\sigma} < Z < \frac{0.0026 - 0.002}{\sigma}\right)$ $= P\left(\frac{-0.0006}{\sigma} < Z < \frac{0.0006}{\sigma}\right).$ Therefore, $P\left(Z < \frac{0.0006}{\sigma}\right) = 0.9975$. Therefore, $\frac{0.0006}{\sigma} = 2.81$ and $\sigma = 0.000214.$ Part i) - 2 marks ii) - 3 marks iii) - 3 marks						
129	**Question:** An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find i). the probability that a bulb burns between 778 and 834 hours. ii). Probability that a bulb has a life of at least 1000 hours iii). What percent of bulbs have their life within ± 2σ from their own mean life, where σ is the standard deviation? **Solution and Scheme:** a. $P(778 < X < 834) \approx 0.5112$ --- 3 marks b. $P(X > 1000) \approx 0.000000287$ (very close to 0) --- 3 marks c. Percent of bulbs within $\pm 2\sigma$ from mean $\approx 95.45\%$ ---2 marks	3	5	Continuous Probability distribution	Moderate	8.00	3

130	**Question:** Assume X is a normally distributed with a mean of 10 and standard deviation of 2. Determine the following: i. P(X<13) ii. P(X>9) iii. P(6<X<14) iv. P(2<X<4) **Solution and Scheme:** i. $P(X < 13) \approx 0.9332$ ii. $P(X > 9) \approx 0.6915$ iii. $P(6 < X < 14) \approx 0.9545$ iv. $P(2 < X < 4) \approx 0.0013$ Each answer 2 marks	3	5	Continuous Probability distribution	Moderate	8.00	3
131	**Question:** BBMP in Bangalore city has installed 2000 electric lamps for street lighting in an area. These lamps have a mean burning life of 1000 hours with a standard deviation of 200 hours. Assume normal distribution is a close approximation to this case. i) What is the probability that a lamp will fail in the first 700 burning hours? ii) What is the probability that a lamp will fail between 900 and 1300 burning hours? iii) What is the probability that a lamp will burn between 899 hours and 901 hours before it fails? iv) After how many burning hours would we expect 10% of the lamps to be left? **Solution and Scheme:** a. $P(X < 700) \approx 0.0668$ b. $P(900 < X < 1300) \approx 0.6247$ c. $P(899 < X < 901) \approx 0.0035$ d. Time after which 10% of lamps are left ≈ 1256.31 hours Each answer 2 marks	3	5	Continuous Probability distribution	Moderate	8.00	3
132	**Question:** The thickness of a conductive coating in micrometers has a density function of 600x-2 for 100 µm < x < 120 µm. i) What is the mean of the coating thickness (in µm). ii) What is the variance of coating thickness (in µm2). iii) If the coating costs \$0.50 per µm of thickness on each part, what is the average cost of the coating per part? (in dollars) **Solution and Scheme:** a. Mean thickness ≈ 109.39 µm --- 3 marks b. Variance ≈ 33.19 µm² --- 3marks c. Average coating cost per part at \$0.50/µm--- 2marks	3	5	Continuous Probability distribution	Moderate	8.00	3
133	**Question:**	3	5	Continuous Probability	Moderate	8.00	3

	The thickness of a flange on an aircraft component is uniformly distributed between 0.95 and 1.05 millimeters. Determine the following: (i) Cumulative distribution function of flange thickness (ii) Proportion of flanges that exceed 1.02 millimeters (iii) Variance of flange thickness (iv) Thickness exceeded by 80% of the flanges **Solution and Scheme:** Let X denote the thickness of flange in millimeters. Then X is uniform over $[0.9, 1.1]$, and its pdf is $$f(x) = \frac{1}{1.1 - 0.9} = 5 \text{ for } 0.9 \le x \le 1.1$$ **a)** $$F(x) = \int_{0.9}^{x} f(t)dt = \int_{0.9}^{x} 5dt = 5(x - 0.9) \text{ for } 0.9 \le x \le 1.1$$ **b)** $$P(X > 1.05) = 1 - P(X \le 1.05) = 1 - 5(1.05 - 0.90) = 1 - 0.75 = \mathbf{0.25}$$ **c)** $$Var(X) = E(X^2) - [E(X)]^2 = \int_{0.9}^{1.1} 5x^2 dx - 1 = \frac{5x^3}{3}\Big|_{0.9}^{1.1} - 1 = \frac{5}{3}\left[(1.1)^3 - (0.9)^3\right] - 1$$ **d)** $$P(x < X) = 0.80 \Rightarrow 1 - P(X \le x) = 0.80$$ $$5(x - 0.9) = 0.2 \Rightarrow \mathbf{x = 0.94}$$ Each answer 2 marks			distribution			
134	**Question:** The compressive strength of samples of cement can be modeled by a normal distribution with a mean of 6000 kilograms per square centimeter and a standard deviation of 100 kilograms per square centimeter. i) What is the probability that a sample's strength is less than 6250 Kg/cm²? ii) What is the probability that a sample's strength is between 5800 and 5900 Kg/cm²? iii) What strength is exceeded by 95% of the samples? **Solution and Scheme:** i. 0.99379 - 2 marks ii. 0.13591 -- 3marks iii. 5835 --- 3marks	3	5	Continuous Probability distribution	Moderate	8.00	3
135	**Question:** The time it takes a cell to divide (called mitosis) is normally distributed with an average time of one hour and a standard deviation of 5 minutes. (i) What is the probability that a cell divides in less than 45 minutes? (ii) What is the probability that it takes a cell more than 65 minutes to divide? (iii) What is the time that it takes approximately 99% of all cells to complete mitosis? **Solution and Scheme:**	3	5	Continuous Probability distribution	Moderate	8.00	3

a) $P(X<45) = P\left(Z < \frac{45-65}{5}\right)$
$= P(Z < -3)$
$= 0.00135$

b) $P(X > 65) = P\left(Z > \frac{65-60}{5}\right)$
$= PZ > 1)$
$= 1 - P(Z < 1)$
$= 1 - 0.841345$
$= 0.158655$

c) $P(X < x) = P\left(Z < \frac{x-60}{5}\right) = 0.99.$

Therefore, $\frac{x-60}{5} = 2.33$ and $x = 71.6$

i) --- 3 marks

ii) --- 3 marks

iii) - 2marks

136	**Question:** A machine fills bottles with a mean volume of 500 ml and a standard deviation of 5 ml. If a random sample of 36 bottles is taken, find the probability that the i) sample mean lies between 498 and 502 ml and ii) sample mean lies between 496 and 504 ml. **Solution and Scheme:** Given: • $\mu = 500, \sigma = 5, n = 36$ • Standard error (SE) = $\frac{5}{\sqrt{36}} = \frac{5}{6} \approx 0.833$ We standardize: $P(498 < \bar{X} < 502) = P\left(\frac{498-500}{0.833} < Z < \frac{502-500}{0.833}\right) = P(-2.4 < Z < 2.4)$ Using standard normal table: $P(Z < 2.4) \approx 0.9918, \quad P(Z < -2.4) \approx 0.0082 \Rightarrow P = 0.9918 - 0.0082 = 0.9836$ So, the probability is approximately **0.984**. For $\bar{X} = 496$, $z = \frac{496-500}{0.8333} = -4.80$ For $\bar{X} = 504$, $z = \frac{504-500}{0.8333} = 4.80$ Thus, $P(496 < \bar{X} < 504) = P(-4.80 < Z < 4.80)$ From the Z-table, $P(Z < 4.80) \approx 0.999999$ $P(Z < -4.80) \approx 0.000001$ Therefore, $P(-4.80 < Z < 4.80) \approx 0.999998$ Each part right answer 4 marks	3	6	Sampling Distribution	Moderate	8.00	3
137	**Question:** Discuss the Central Limit Theorem (CLT). State the conditions under which it is applicable and discuss any four important applications/significance of the Central Limit Theorem in statistical inference. **Solution and Scheme:**	3	5	Continuous Probability distribution	Moderate	8.00	3

	The Central Limit Theorem states that for a large enough sample size (n), the sampling distribution of the sample mean will be approximately normally distributed, regardless of the original population distribution, provided the population has a finite mean and variance. --- 3 marks **Conditions for Applying the CLT** 1. Samples are selected randomly. 2. Sample observations are independent. 3. The population has a finite mean and finite variance. 4. The sample size is sufficiently large (generally n≥30n \geq 30n≥30). For normally distributed populations, smaller sample sizes may also be adequate ---- 2 marks **Significance / Applications of the Central Limit Theorem** 1. Approximates the sampling distribution by a normal distribution 2. Foundation for Statistical Inference 3. Construction of Confidence Intervals 4. Hypothesis Testing 5. Quality Control 6. Decision Making ---- 3 marks						
138	**Question:** A component has an average lifetime of 5 hours. Assuming exponential distribution, find the probability that the component lasts more than 8 hours. **Solution and Scheme:** Given: Mean = 5 → λ = 1/5 We want: $P(X > 8) = e^{-\lambda x} = e^{-(1/5)\cdot 8} = e^{-1.6} \approx 0.2019$ So, the probability is approximately **0.202**.	3	6	Sampling Distribution	Moderate	2.00	2
139	**Question:** The life of a semiconductor laser at a constant power is normally distributed with a mean of 7000 hours and a standard deviation of 600 hours. i) What is the probability that a laser fails before 6000 hours? ii) What is the life in hours that 95% of the lasers exceed? iii) If three lasers are used in a product and they are assumed to fail independently, what is the probability that all three are still operating after 7000 hours? **Solution and Scheme:** Let X denote the life (in hours) of this semiconductor laser. Then X is normal with $\mu = 7000$ and $\sigma = 600$. **a)** $$P(X < 6000) = P\left(Z \le \frac{6000 - 7000}{600}\right) = P(Z \le -1.67) = \mathbf{0.047}$$ **b)** Find x such that $P(X > x) = 0.95$ or equivalently $P(X \le x) = 0.05$. $$P(X \le x) = P\left(Z \le \frac{x - 7000}{600}\right) = 0.05 \Rightarrow \frac{x - 7000}{600} = -1.65$$ Therefore, $\mathbf{x = 6010}$. **c)** If we define $p = P(X > 7000) = 0.5$, then by independence of the failures $$P(\text{All 3 lasers still operate}) = p^3 = \mathbf{1/8}$$ i) - 3 marks ii) -- 3 marks iii) -- 2marks	3	6	Sampling Distribution	Moderate	8.00	3
140	**Question:** The summarized data on 10 observations are given below. a. Estimate β1 and βo assuming positive correlation between X and Y. b. Obtain an estimate of the underlying variance. c. What is the proportion of variability explained by the regression model?	4	7	Simple linear Regression and Correlation	Moderate	8.00	3

d. Assuming normal distribution of the errors, how will a residual plot of errors vs predicted values appear in this case? (Show an indicative plot).

$$\sum_i (y_i - \hat{y}_i)^2 = 3{\cdot}267, \quad \sum_i y_i = 63{\cdot}3, \quad \sum_i y_i^2 = 423{\cdot}49, \quad \sum_i x_i = 139, \quad \sum_i x_i^2 = 2239$$

Solution and Scheme:

$\beta_1 \approx 0.7176$ --- 1 mark

$\beta_0 \approx 1.406$ ---- 1 mark

$\sigma^2 \approx 0.4071$ ---- 2 marks

$R^2 \approx 0.9829$ ----- 2 marks

y= 1.406 +0.7176x ---- 1 mark

Plot - 1 mark

141	**Question:** In an oil refinery, chemical reactions are observed for a set of 20 pairs of values (xi, yi), and the relevant quantities are derived as: $\sum x_i^2 = 130000$, $\sum y_i^2 = 88.6$, $\sum x_i y_i = 1500$, $\sum x_i = 1500$, $\sum y_i = 10$ i) Compute the intercept and slope parameters and write the estimated linear regression model relating x and y. ii) Compute the predicted value y for an input x = 85, and find the corresponding residual if the observed value is 0.8 iii) Estimate the error variance and the proportion of variability not explained by the model. **Solution and Scheme:** y = -2.7143+ 0.042857x ---- 4 marks Ycap = $-2.7143+0.04286 \cdot 85 = -2.7143+3.6431=0.9288$ --- 1 mark Residual=$y_{obs} - \hat{y} = 0.8 - 0.9288 = -0.1288$ -- 1 mark 32 % of variance not explained by the model --- 2 marks	4	7	Simple linear Regression and Correlation	Moderate	8.00	3
142	**Question:** During the fall harvest season, pumpkins are sold in large quantities. Often, instead of weighing the pumpkins prior to sale, the operator will just place the pumpkin in the appropriate circular cutout on the counter. When asked why this was done, one farmer replied, I can tell the weight of the pumpkin from its circumference. To determine whether this was really true, a sample of 23 pumpkins were measured for circumference and weighed and the data is provided below	4	7	Simple linear Regression and Correlation	Hard	16.00	3

Circumference (cm)	Weight (gms)	Circumference (cm)	Weight (gms)	Circumference (cm)	Weight (gms)
50	1200	57	2000	47	1400
55	2000	66	2500	51	1500
54	1500	82	4600	63	2500
52	1700	83	4600	33	500
37	500	70	3100	43	1000
52	1000	34	600	50	1500
53	1500	51	1500	49	1600

60	2300	59	2100		

i. Assuming a linear relationship, use the least-squares method to compute the regression coefficients b0 and b1.

ii. Interpret the meaning of the slope, b1, in this problem.

iii. Predict the mean weight for a pumpkin that is 60 centimeters in circumference.

iv. Do you think it is a good idea for the farmer to sell pumpkins by circumference instead of weight? Explain.

Solution and Scheme:

a. Regression coefficients:

- b0=−2637.24
- b1=82.55 ---- 8 marks

b. Interpretation of slope b1:
Each 1 cm increase in pumpkin circumference is associated with an average increase of **82.55 grams** in weight. --- 2 mark

c. Predicted weight for a 60 cm circumference:
Approximately **2315.95 grams --- 2 marks**

d. Is it a good idea to sell by circumference?
Yes, if the linear relationship is strong (as it appears), using circumference as a proxy for weight can be practical and efficient. However, accuracy may vary for outliers or irregular shapes, so calibration or verification is recommended. -- 4 marks

143	**Question:**	4	7	Simple linear Regression and Correlation	Easy	8.00	2

An online retailer wants to predict customer spending based on the number of website visits. Explain how linear regression can be applied in this scenario. Discuss the advantages and limitations of using linear regression for prediction.

Solution and Scheme:

Linear regression is a statistical technique used to establish a relationship between an independent variable and a dependent variable. In this case:

- **Independent variable (X):** Number of website visits
- **Dependent variable (Y):** Customer spending

Procedure:

1. Collect historical data on website visits and customer spending.
2. Plot the data using a scatter diagram.
3. Estimate the regression equation.
4. Validate the model using statistical measures such as (R^2).
5. Use the equation to predict future customer spending.

Advantages

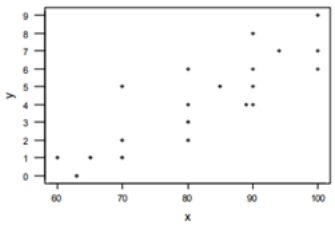
1. Simple to understand and implement.
2. Helps predict future customer spending.
3. Identifies the relationship between variables.
4. Assists in marketing and sales planning.
5. Supports data-driven business decisions.

Limitations

1. Assumes a linear relationship between variables.
2. Sensitive to outliers.

3. Cannot establish causation.
4. Accuracy decreases if assumptions are violated.
5. Poor predictor when important variables are omitted.

2 marks each part

144	**Question:** Data for an increase in blood pressure, y (in mm of Hg) versus sound level exposure, x (in decibels) are given below for 20 instances: x: 60, 63, 65, 70, 70, 70, 80, 90, 80, 80 y: 1, 0, 1, 2, 5, 1, 4, 6, 2, 3 x: 85, 89, 90, 90, 90, 90, 90, 100, 100, 100 y: 5, 4, 6, 8, 4, 5, 7, 9, 7, 6 (i) Draw a scatter diagram of y v/s x. Does a simple linear regression model seem reasonable? (ii) Fit the simple linear regression model using least squares. Find an estimate of σ^2 (iii) Find the predicted blood pressure at a sound level of 85dB. What is the residual in this case? (iv) Find the predicted blood pressure at a sound level of 75dB and 105dB **Solution and Scheme:** a) Yes, a linear regression would seem appropriate, but one or two points appear to be outliers.  ---- 4 marks b) $\hat{\sigma}^2 = 1.9818$ and $\hat{y} = -9.8131 + 0.171484x$ c) $\hat{y} = 4.76301$ at $x = 85$ -----8 marks d) 3.0479 at x=75 and 8.1923 at x=105db ---4 marks	4	7	Simple linear Regression and Correlation	Moderate	16.00	3
145	**Question:** For a set of 20 pairs of values (x_i,y_i), $\sum x_i^2 = 29.29$, $\sum y_i^2 = 170044.53$, $\sum x_i y_i = 2214.66$, $\sum x_i = 23.92$, $\sum y_i = 1843.21$. Write the regression line equation Subsequently it was found that the values for the pair $(x_{20}, y_{20}) = (2, 90)$ was actually wrong and the correct pair $(x_{20}, y_{20}) = (0, 60)$ was inserted in the set after deleting the wrong pair. Write the new regression line equation. What is the residual at x = 0 with the corrected data set? **Solution and Scheme:** • **Old regression line - 2 marks** • **New regression line equation:** y^=49.63+37.44x ---- 5 marks • **Residual at x=0** 10.37 --- 1mark	4	7	Simple linear Regression and Correlation	Moderate	8.00	3
146	**Question:**	4	7	Simple linear	Moderate	16.00	3

The Store in charge of a Big Bazaar would like to use shelf space to predict the sales of certain type of food item. A random sample of 12 equal-sized stores is selected, with the following results

Store	Shelf Space (X) (Feet)	Weekly Sales (Y) ($)
1	5	160
2	5	220
3	5	140
4	10	190
5	10	240
6	10	260
7	15	230
8	15	270
9	15	280
10	20	260
11	20	290
12	20	310

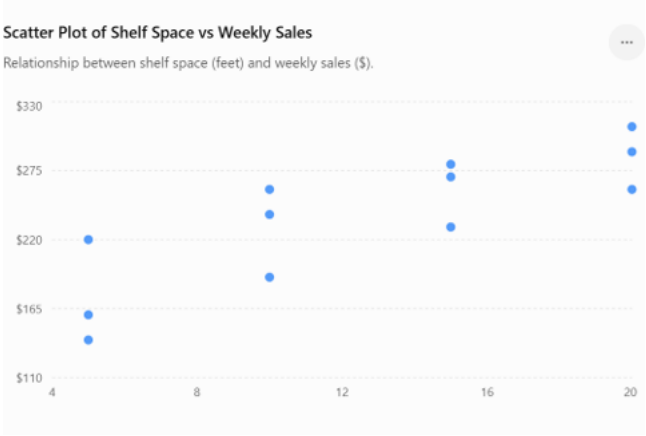
i. Construct a scatter plot.

ii. Calculate the slope and Intercept value.

iii. Interpret the meaning of the slope, b1, in this problem.

iv. Predict the mean weekly sales (in hundreds of dollars) of food item for stores with 8 feet of shelf space.

Solution and Scheme:



--- 4 marks

b. Regression Line Equation:

y^=145.0+7.4x --- 8 marks

c. Interpretation of the slope
Each additional foot of shelf space is associated with an average increase of **$7.40** in weekly sales. --- 2 marks

d. Predicted mean weekly sales for 8 feet of shelf space:
$204.20 --- 2 marks

Regression and Correlation

147 **Question:**

A physicist wishes to use the model $y = k/x^r$ where x = independent variable and y = dependent variable. The following data is observed. Use the transformation ln y =ln k – r ln x, and use least square estimators of linear regression.

x_i	50	60	70	80	90
y_i	64.5	51.0	40.6	33.4	26.0

i) Obtain estimates of k and r.

ii) What is the predicted value of y at x =60 and the corresponding residual.

iii) Provide an estimate for the underlying variance of the yi values.

Solution and Scheme:

r≈2.16, k≈780.0 --- 8 marks

y prediction = 50.2 --- 2 marks

4 7 Simple linear Regression and Correlation Hard 16.00 3

	Residual at x=60x = 60: Residual=yobserved−ypredicted=51.0−50.2=0.8 --- 2marks c) Estimate of variance: $\sigma^2 \approx \sum(yi-ypred)2/n-2 \approx 1.78$ --- 4 marks						
148	**Question:** Regression methods were used to analyze the data from a study investigating the relationship between roadway surface temperature (x) and pavement deflection (y). Summary quantities were n = 20. $\sum y_i=12.75$, $\sum y_i^2=8.86$, $\sum x_i=1478$, $\sum x_i^2=143{,}215.8$, and $\sum x_i y_i=1083.67$ a. Calculate the least squares estimates of the slope and intercept. Graph the regression line. Estimate σ^2. b. Use the equation of the fitted line to predict what pavement deflection would be observed when the surface temperature is 85°F. c. What is the mean pavement deflection when the surface temperature is 90°F? d. What change in mean pavement deflection would be expected for a 1°F change in surface temperature? **Solution and Scheme:** Ycap = 0.32999+0.00416x --- 4 marks Graph - 4 marks Variance =0.00796 --- 2 marks Ycap at 85F = 0.6836 ---2 marks Ycap at 90F = 0.7044 --- 2 marks d. 0.00416 --- 2 marks	4	7	Simple linear Regression and Correlation	Moderate	16.00	3
149	**Question:** An article in Concrete Research [“Near Surface Characteristics of Concrete: Intrinsic Permeability” (1989, Vol. 41)] presented data on compressive strength x and intrinsic permeability y of various concrete mixes and cures. Summary quantities are n = 14, $\sum y_i=572$, $\sum y_i^2=23{,}530$, $\sum x_i=43$, $\sum x_i^2=157.42$ and $\sum x_i y_i=1697.80$ Assume that the two variables are related according to the simple linear regression model. i. Calculate the least squares estimates of the slope and intercept. Estimate σ^2. Graph the regression line. ii. Use the equation of the fitted line to predict what permeability would be observed when the compressive strength is x = 4.3. iii. Give a point estimate of the mean permeability when compressive strength is x = 3.7. iv.. Suppose that the observed value of permeability at x = 3.7 is y = 46.1. Calculate the value of the corresponding residual. **Solution and Scheme:**	4	7	Simple linear Regression and Correlation	Moderate	8.00	3

	a) $y_i = \beta_0 + \beta_1 x_1 + \varepsilon_i$ $S_{xx} = 157.42 - \frac{43^2}{14}$ $= 25.348571$ $S_{xy} = 1697.80 - \frac{43(572)}{14}$ $= -59.057143$ $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-59.057143}{25.348571} = -2.330$ $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{572}{14} - (-2.3298017)(\frac{43}{14}) = 48.013$ b) $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ $\hat{y} = 48.012962 - 2.3298017(4.3) = 37.99$ c) $\hat{y} = 48.012962 - 2.3298017(3.7) = 39.39$ d) $e = y - \hat{y} = 46.1 - 39.39 = 6.71$ i) - 5marks ii)- 1 mark iii)-1 mark iv) --1 marks						
150	**Question:** State the assumptions of linear regression model? **Solution and Scheme:** • **Linearity**: The relationship between independent and dependent variables is linear. • **Independence**: Observations are independent of each other. • **Homoscedasticity**: Constant variance of residuals across all levels of the independent variable. • **Normality**: Residuals follow a normal distribution. • **No Multicollinearity**: Independent variables are not highly correlated with each other. • **No Autocorrelation**: Residuals are not correlated with each other (especially important for time series data).	4	7	Simple linear Regression and Correlation	Easy	2.00	3
151	**Question:** In a maintenance department of a firm, the life of an electronic sensor equipment is found to be normally distributed. A random sample of 30 sensors has a sample mean life of 800 hours, and a sample standard deviation of 70 hours. (i) Construct a 95% two-sided confidence interval for the mean life of the sensors. (ii) Construct a 99% two-sided confidence interval for the variance of life of sensors. (iii) If the population standard deviation of the life of sensors is known to be 90 hours, construct a 99% two –sided confidence interval for mean life of the sensors. (iv) Comment on difference in answers to (ii) & (iii) **Solution and Scheme:** i) CI=(773.9,826.1) -- 2M ii) CI=(2712.4,10835.1) --- 2M ii)CI=(757.7,842.3) --- 2M iv) Comment on Differences: The confidence interval for variance (ii) is much wider due to the chi-square distribution being highly skewed, especially at small sample sizes. The confidence interval for the mean in (iii) is narrower than in (i) because using the known population standard deviation results in a more precise estimate (z-distribution vs. t-distribution).	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	8.00	3

2M

152	**Question:** For the sample (6, 3, 2, 1, 1, 0, 1, 3, 3, 2) from a population representing the number of defects in returned appliances by customers, i) Construct a 90 % confidence interval on population mean no of defects per returned appliance, if the population standard deviation is known to be 2 defects. ii) What should the sample size be in order to keep the width of the confidence interval on mean to less than 2? iii) Construct a 95 % confidence interval on population mean assuming that the population variance is unknown. **Solution and Scheme:** a) CI=2.2±1.04 CI=(1.16,3.24) ---------- 3 marks b) n=3.29⇒n≈11 ---------- 3 marks c)CI=2.2±1.27 CI=(0.93,3.47) ---------- 2 marks	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	8.00	3
153	**Question:** 10 values denoting the number of failed specimens per day in a load test, collected at random on 10 different days are: 100, 115, 150, 170, 90, 135, 180, 110, 150, 155. Assuming that the failed specimens follow normal distribution approximately, test if the mean number of failed specimens is at most 120, at 10% significance **Solution and Scheme:** **Hypothesis Testing:** • Null Hypothesis: H0:μ≤120 • Alternative Hypothesis: H1:μ>120 • Significance level: α=0.10 • Sample size: n=10 ------- 2 marks **Step 1: Compute sample mean and standard deviation** $$x = \frac{100 + 115 + 150 + 170 + 90 + 135 + 180 + 110 + 150 + 155}{10} = 135.5$$ $$s = \sqrt{\frac{\sum (x_i - x)^2}{n - 1}} = 30.76$$ ---------- 2 marks	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	8.00	3

Step 2: Compute t-statistic

$$t = \frac{\bar{x} - 120}{s/\sqrt{n}}$$

$$t = \frac{135.5 - 120}{30.76/\sqrt{10}}$$

$$t = \frac{15.5}{9.73} = 1.59$$

Step 3: Critical t-value at $\alpha = 0.10$ **for** $df = 9$

$$t_{0.10,9} \approx 1.383$$

Step 4: Decision Since $t = 1.59 > 1.383$, we **reject** H_0 at 10% significance level.
Conclusion: There is enough evidence to suggest that the mean number of failed specimens is greater than 120.

---- 4marks

154	**Question:**	5	8	Statistical Inference for a single sample, Interval Estimation	Easy	8.00	3

The number of defects in printed circuit boards is hypothesized to follow a Poisson distribution. A random sample of n = 60 printed circuit boards has been collected, and the following number of defects observed. Perform a test of hypothesis.

No. of Defects	Observed Frequency
0	32
1	15
2	9
3	4

Solution and Scheme:

The variable of interest is the form of the distribution of defects in the printed circuit boards

H_o: The form of the distribution of defects is Poisson

H_1: The form of the distribution of defects is not Poisson

α = 0.05 ---------- 1 marks

The estimate of the mean number of defects per board is the sample average, that is,

Sample Average = (32.0 + 15.1 + 9.2 + 4.3) / 60 = 0.75. ---- 1 marks

The expected frequencies are computed by multiplying the sample size n = 60 times the probabilities pi. That is, Ei = npi. The expected frequencies follow:

No. of Defects	Probability	Expected Frequency
0	0.472	28.32
1	0.354	21.24
2	0.133	7.98
3 (or more)	0.041	2.46

Because the expected frequency in the last cell is less than 3, combine the last two cells

No. of Defects	Observed Frequency	Expected Frequency
0	32	28.32
1	15	21.24
2 (or more)	13	10.44

---------- 4 marks

From table: Chi square = 3.84, cannot reject null hypothesis ---- 2 marks

155	**Question:** Explain the steps involved in conducting a hypothesis testing with a suitable example. **Solution and Scheme:** Seven – step Procedure: 1. Parameter of interest: The parameter of interest is μ, the mean burning rate. 2. Null hypothesis: H0: μ = 50 centimeters per second 3. Alternative hypothesis: H1: μ ≠ 50 centimeters per second 4. Test statistic: The test statistic is 5. Reject H0 if: Reject H0 if the P-value is less than 0.05. To use a fixed significance level test, the boundaries of the critical region would be $z_{0.025} = 1.96$ and $-z_{0.025} = -1.96$. 6. Computations: Because $\bar{x} = 51.3$ and $\sigma = 2$, 7. Conclusion: Because the P-value $= 2[1 - \Phi(3.25)] = 0.0012$, we reject H0: μ = 50 at the 0.05 level of significance. 8. Interpretation: The mean burning rate differs from 50 centimeters per second, based on a sample of 25 measurements. In fact, there is strong evidence that the mean burning rate exceeds 50 centimeters per second. Each step - 1 mark	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	8.00	2
156	**Question:** a) What are the hypothesis statements and the statistic to be used in a two-sided test of variance of a population? What are the assumptions to use this statistic and what is the rejection region for this test? b) State the probability of a t-random variable with 25 degrees of freedom taking on any value between 2.06 and 3.45 c) What is the value x such that a chi-square random variable with 15 degrees of freedom takes on at the most a value of x with 95% likelihood? **Solution and Scheme:** Hypothesis statements - 2 marks Probability of a t-Random Variable (df = 25) Using a t-distribution table or calculator: • $P(t<3.45) \approx 0.9985$ • $P(t<2.06) \approx 0.975$ • $P(2.06<T<3.45) = 0.999-0.97 = 0.022$ --- 3marks **Chi-Square Value for 95% Cumulative Probability (df = 15)** You are asked to find the value xxx such that: $P(\chi 2 \leq x) = 0.95$ Using a chi-square table ≈24.996 --- 3marks	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	8.00	4
157	**Question:** Length L of certain specialty parts is normally distributed with variance of 25 mm^2. If a customer specifies his requirement on length as $100^{\pm 2.00}$ mm and rejects parts outside these limits, what is the probability of a Type I error in a test with H_0: μ = 100 mm versus H_1: μ ≠ 100 mm ? Sample size, n = 50 parts. **Solution and Scheme:** SE≈0.707 --- 2 M Now, calculating the z-scores for the rejection limits: z=−2.83 - 2 marks z=2.83 From standard normal tables, P(Z<−2.83)≈0.0023 --- 2 marks	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	8.00	4

	Thus, the probability of a Type I error: $\alpha=P(Z<-2.83)+P(Z>2.83)=0.0023+0.0023=0.0046$ --- 2 M So, the probability of Type I error is **0.46%**.						
158	**Question:** Test at 10 % significance if the variance of a population is greater than 10, with the help of sample data 24, 22, 25, 23, 20, 23, 21, 22, 25, 22 **Solution and Scheme:** Mean = 22.7 --- 1 M Variance = 3.88 ----- 2 M Chi square =3.49 ------ 3 M Chi square table - 14.68 --- 1 M Since $\chi 2=3.49$ is **less than** 14.68, we **fail to reject** H0. **Conclusion:** There is **not enough evidence** to conclude that the population variance is greater than 10 at the 10% significance level. ---- 1 M	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	8.00	4
159	**Question:** A 1992 article in the Journal of the American Medical Association ("A Critical Appraisal of 98.6 Degrees F, the Upper Limit of the Normal Body Temperature, and Other Legacies of Carl Reinhold August Wundrlich") reported body temperature, gender, and heart rate for a number of subjects. The body temperatures for 25 female subjects follow: 97.8, 97.2, 97.4, 97.6, 97.8, 97.9, 98.0, 98.0, 98.0, 98.1, 98.2, 98.3, 98.3, 98.4, 98.4, 98.4, 98.5, 98.6, 98.6, 98.7, 98.8, 98.8, 98.9, 98.9, and 99.0. Test the hypotheses H_0: $\mu = 98.6$ versus using H_1:$\mu \neq 98.6$ using $\alpha = 0.05$. **Solution and Scheme:** t= -1.79 --- 4 marks $t_{0.025,24}\approx\pm2.064$ -- 2 mark Since |t|= 1.79 is **less than** 2.064, **fail to reject** H0. No significant difference in10 mean body temperature at $\alpha=0.05$. --2 M	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	8.00	4
160	**Question:** The sample data 24, 22, 25, 23, 20, 23, 21, 22, 25, 22 was collected by a student during an experiment. i) What is the estimate of the population mean, population variance as well as variance of the sample mean of all such samples of size 10? ii) What is a 90% lower bound for the population mean of the above data? iii) What is the 95 % upper bound for the population variance of the above data? **Solution and Scheme:**	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	8.00	4

(a) Estimates:

$$\bar{x} = \frac{\sum x_i}{10} = \frac{24+22+25+23+20+23+21+22+25+22}{10} = 22.7$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = 3.88$$

$$\text{Variance of Sample Mean} = \frac{s^2}{n} = \frac{3.88}{10} = 0.388$$

(b) 90% Lower Bound for Population Mean:

$$LB = \bar{x} - t_{\alpha, n-1} \times \frac{s}{\sqrt{n}}$$

For $t_{0.10,9} \approx 1.383$:

$$LB = 22.7 - (1.383 \times \frac{1.97}{\sqrt{10}})$$

$$LB = 22.7 - 0.86 = 21.84$$

(c) 95% Upper Bound for Population Variance:

$$UB = \frac{(n-1)s^2}{\chi^2_{\alpha, n-1}}$$

For $\chi^2_{0.05,9} \approx 16.92$:

$$UB = \frac{9 \times 3.88}{16.92} = 2.06$$

i) - 2 marks

ii) -- 3marks

iii) -- 3 marks

161	**Question:** An engineer who is studying the tensile strength of a steel alloy intended for use in golf club shafts knows that tensile strength is approximately normally distributed . A random sample of 12 specimens has a mean tensile strength of 3250 psi. (i) Test the hypothesis that mean strength is 3500 psi. Use 1 % significance level. Standard deviation is known to be 60 psi (ii) What is the smallest level of significance at which you would be willing to reject the null hypothesis in part (a)? (iii) Explain how you could answer the question in part (a) with a two-sided confidence interval on mean tensile strength. (iv) How will the test be performed if the standard deviation is not known? **Solution and Scheme:**	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	8.00	4

(a) **Hypothesis Test:**

$$t = \frac{3250 - 3500}{60/\sqrt{12}} = \frac{-250}{17.32} \approx -14.43$$

Critical value at $\alpha = 0.01$, $t_{0.005,11} \approx \pm 3.106$. Since $|t| = 14.43 > 3.106$, **reject** H_0.

(b) **Smallest Significance Level:** p-value for $t = -14.43$ is **essentially zero**, so we would reject H_0 at any reasonable significance level.

(c) **Two-Sided Confidence Interval:**

$$CI = x \pm t_{\alpha/2,n-1} \times \frac{\sigma}{\sqrt{n}}$$

For $t_{0.005,11} \approx 3.106$:

$$CI = 3250 \pm (3.106 \times 17.32)$$

$$CI = 3250 \pm 53.8$$

$$CI = (3196.2, 3303.8)$$

Since 3500 is **outside** this interval, we reject H_0.

(d) **If Standard Deviation is Unknown:** Replace σ with sample standard deviation s and use the t-distribution instead of the normal distribution.

Each sub answer - 2 marks

162	**Question:** The sugar content of the syrup in canned peaches is normally distributed. A random sample of n = 10 cans yields a sample standard deviation of s = 4.8 milligrams. i) Find a 90% two-sided confidence interval for population standard deviation. ii) Find a 95% upper bound and a 95% lower bound for population standard deviation. How do these bounds compare with the limits of part (a)?	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	8.00	5

Solution and Scheme:

(a) 90% Two-Sided Confidence Interval for Population Standard Deviation Using chi-square distribution:

$$CI = \left(\sqrt{\frac{(n-1)s^2}{\chi^2_{upper}}}, \sqrt{\frac{(n-1)s^2}{\chi^2_{lower}}} \right)$$

For $n = 10$, $s = 4.8$, $\chi^2_{0.95,9} \approx 16.92$, $\chi^2_{0.05,9} \approx 3.33$:

$$CI = \left(\sqrt{\frac{9 \times (4.8)^2}{16.92}}, \sqrt{\frac{9 \times (4.8)^2}{3.33}} \right)$$

$$CI = \left(\sqrt{\frac{207.36}{16.92}}, \sqrt{\frac{207.36}{3.33}} \right)$$

$$CI = (3.52, 7.86)$$

(b) 95% Upper and Lower Bounds for Population Standard Deviation Using chi-square distribution for upper bound:

$$UB = \sqrt{\frac{(n-1)s^2}{\chi^2_{0.05,9}}}$$

For $\chi^2_{0.05,9} \approx 16.92$:

$$UB = \sqrt{\frac{207.36}{16.92}} = 3.52$$

Using chi-square distribution for lower bound:

$$LB = \sqrt{\frac{(n-1)s^2}{\chi^2_{0.95,9}}}$$

For $\chi^2_{0.95,9} \approx 2.70$:

$$LB = \sqrt{\frac{207.36}{2.70}} = 8.76$$

Comparison:

- The 90% confidence interval provides both upper and lower limits that are narrower than the 95% bounds.
- The 95% upper bound is the same as the lower limit from the 90% confidence interval.
- The 95% lower bound is **wider** than the upper limit from the 90% confidence interval due to the smaller chi-square value.

Each sub part - 4 marks

163	**Question:** Cloud seeding has been studied for many decades as a weather modification. The rainfall in acre-feet from 20 clouds that were selected at random and seeded with silver nitrate are as follows: 18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9, 31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, and 31.6. (i) construct a 95 % one-sided lower bound on the mean diameter. (ii) construct a 90 % two-sided confidence interval on the mean diameter. What is the width of this interval? (iii) If the width of the interval in part (b) were to be halved, how much should be the sample size? Use trial and error method if needed. **Solution and Scheme:**	5	8	Statistical Inference for a single sample, Interval Estimation	Hard	8.00	4

(a) 95% One-Sided Lower Bound on Mean Rainfall Given $n = 20$, calculate sample mean $\bar{x}$ and standard deviation s:

$$\bar{x} = \frac{\sum x_i}{20} = 26.51, \quad s = 4.78$$

Lower bound formula:

$$LB = \bar{x} - t_{\alpha, n-1} \times \frac{s}{\sqrt{n}}$$

For $t_{0.05,19} \approx 1.729$:

$$LB = 26.51 - (1.729 \times \frac{4.78}{\sqrt{20}})$$

$$LB = 26.51 - 1.85 = 24.66$$

(b) 90% Two-Sided Confidence Interval on Mean Rainfall

$$CI = \bar{x} \pm t_{\alpha/2, n-1} \times \frac{s}{\sqrt{n}}$$

For $t_{0.05,19} \approx 1.729$:

$$CI = 26.51 \pm (1.729 \times \frac{4.78}{\sqrt{20}})$$

$$CI = (24.66, 28.36)$$

Width of the interval:

$$\text{Width} = 28.36 - 24.66 = 3.70$$

(c) Required Sample Size to Halve the Width To halve the width $W' = 3.70/2 = 1.85$:

$$W' = 2 \times t_{\alpha/2, n-1} \times \frac{s}{\sqrt{n'}}$$

Solving for n':

$$\sqrt{n'} = \frac{1.729 \times 4.78}{1.85} \approx 4.47$$

$$n' \approx 20$$

Thus, the required sample size should be approximately **40** to halve the width.

i) and ii) --- 5 M

iii) - 3 M

164	**Question:** Amino acid concentrations in 8 packs of dog feeds were recorded in mg/kg of dog feed as follows: 22, 24, 20, 26, 27, 24, 26, 23 i) Perform a test at 5% significance to see if the population variance is greater than 5.5 (mg/kg)2. What is the p-value for this test? ii) Construct a 90 % upper and a 90 % lower confidence bounds for the population variance **Solution and Scheme:**	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	8.00	4

Step 1: Compute Sample Variance

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Mean:

$$\bar{x} = \frac{22+24+20+26+27+24+26+23}{8} = 24$$

Sample variance:

$$s^2 = \frac{(22-24)^2 + (24-24)^2 + (20-24)^2 + (26-24)^2 + (27-24)^2 + (24-24)^2 + (26-24}{7}$$

Step 2: Compute Chi-Square Statistic

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

$$\chi^2 = \frac{7 \times 6.57}{5.5} = \frac{46.0}{5.5} = 8.36$$

Step 3: Determine p-value

For $\chi^2_{0.05,7} \approx 14.07$, since $8.36 < 14.07$, we **fail to reject** H_0. Using a chi-square table, the p-value corresponds to the upper tail probability for $\chi^2 = 8.36$, approximately **0.30**.

Conclusion: There is not enough evidence at $\alpha = 0.05$ to conclude that the population variance is greater than 5.5.

(b) 90% Upper and Lower Confidence Bounds for Population Variance Using chi-square confidence limits:

$$LB = \frac{(n-1)s^2}{\chi^2_{upper}}, \quad UB = \frac{(n-1)s^2}{\chi^2_{lower}}$$

For $\chi^2_{0.95,7} \approx 2.17$ and $\chi^2_{0.05,7} \approx 14.07$:

$$LB = \frac{7 \times 6.57}{14.07} = \frac{46.0}{14.07} = 3.27$$

$$UB = \frac{7 \times 6.57}{2.17} = \frac{46.0}{2.17} = 21.20$$

sub part i) - 5 marks

ii) - 3 marks

165	**Question:** State the addition rule for two events in probability. **Solution and Scheme:** The addition rule for two events A and B is: P(A ∪ B) = P(A) + P(B) – P(A ∩ B)	1	2	Concepts of Probability	Easy	2.00	2
166	**Question:** What is conditional probability? **Solution and Scheme:** Conditional probability is the probability of an event A occurring given that event B has occurred. It is denoted as P(A|B) and calculated using: P(A|B) = P(A ∩ B) / P(B), provided P(B) > 0.	1	2	Concepts of Probability	Easy	2.00	2

168	**Question:** Differentiate between discrete and continuous random variables with examples. **Solution and Scheme:** A discrete random variable takes on countable values (e.g., number of children). A continuous random variable assumes any value within a range (e.g., height).	1	2	Concepts of Probability	Easy	2.00	4
169	**Question:** Define the sampling distribution of the sample mean. **Solution and Scheme:** It is the probability distribution of all possible sample means of a given size from a population. It has a mean equal to the population mean (μ) and a standard deviation equal to the population standard deviation (σ) divided by the square root of the sample size (n).	3	6	Sampling Distribution	Easy	2.00	1
170	**Question:** Define the correlation coefficient and its range. **Solution and Scheme:** The correlation coefficient (r) measures the strength and direction of a linear relationship between two variables. Its value ranges from -1 (perfect negative correlation) to +1 (perfect positive correlation), with 0 indicating no linear correlation.	4	7	Simple linear Regression and Correlation	Easy	2.00	2
171	**Question:** When is a z-test used for population mean inference? **Solution and Scheme:** When the population variance is known and sample size is large (n > 30) or population is normal.	5	8	Statistical Inference for a single sample, Interval Estimation	Easy	2.00	2
172	**Question:** Compare and contrast histograms, box plots, and stem-and-leaf plots in terms of their use with suitable examples **Solution and Scheme:** • **Histograms** show the frequency distribution of continuous data. Useful to see shape, spread, and modality. • **Box plots** summarize data using five-number summary (min, Q1, median, Q3, max) and identify outliers. • **Stem-and-leaf plots** retain the original data values and show distribution shape for small datasets. **Comparison:** • Box plots are best for comparing multiple datasets. • Histograms better show detailed distribution shape. • Stem-and-leaf plots are useful for exploratory analysis of small samples.	1	1	Data types and Charts	Moderate	8.00	2
173	**Question:** What are the assumptions and applications of the binomial and Poisson distributions? Compare the two distributions with suitable examples **Solution and Scheme:**	2	4	Discrete Probability distribution	Moderate	8.00	4

	Binomial Distribution: • Fixed number of trials (n), two outcomes (success/failure), constant success probability (p), independent trials. • Application: Number of defective items in a batch. **Poisson Distribution:** • Models count of events in a fixed interval of time or space, events occur independently, at a constant rate (λ). • Application: Number of emails received per hour. **Comparison:** • Binomial is for a fixed number of trials; Poisson is for events in continuous time/space. • Poisson is often a limiting case of binomial when $n \to \infty$ and $p \to 0$, such that $np = \lambda$. Assumptions - 3 marks Applications - 3 marks Comparison - 2 marks						
174	**Question:** What are the assumptions of simple linear regression? How do you validate them using residual plots? **Solution and Scheme:** **Assumptions:** 1. Linearity 2. Independence of errors 3. Homoscedasticity (constant variance of errors) 4. Normality of errors -- 4 marks **Validation using residual plots:** • Plot residuals vs fitted values: ◦ Random scatter suggests linearity and constant variance. ◦ Patterns indicate non-linearity or heteroscedasticity. • Histogram or Q-Q plot of residuals checks for normality. • Time-sequence plot of residuals checks independence. ---4 marks	4	7	Simple linear Regression and Correlation	Moderate	8.00	2
175	**Question:** Describe the characteristics and applications of the Normal distribution. **Solution and Scheme:** **Characteristics:** • Bell-shaped, symmetric about the mean. • Mean = Median = Mode. • Defined by two parameters: mean (μ) and standard deviation (σ). • Total area under the curve = 1 --- 4marks • Applications --- 4 marks	3	5	Continuous Probability distribution	Easy	8.00	2
176	**Question:**	3	6	Sampling Distribution	Moderate	8.00	2

	What are the key differences between population and sample? Why is sampling important in statistics? **Solution and Scheme:** **Population:** Entire group of interest (e.g., all students in a college). **Sample:** Subset of the population used for analysis (e.g., 100 randomly selected students). **Key Differences:** • Size: Population is large, sample is smaller. • Parameters vs. statistics: Population has fixed parameters (μ, σ); samples have estimators ($\bar{x}$, s). **---- 5 marks** **Importance of Sampling:** • Saves time and cost. • Enables inference about the population using statistical methods. • Often necessary when population data is inaccessible or infinite. -- 3 marks						
177	**Question:** Describe the method of least squares in simple linear regression. How is the best-fit line determined? **Solution and Scheme:** **Method of Least Squares:** Minimizes the sum of squared differences between observed and predicted values. **Model:** y = a + bx Where: • **b** = slope = Cov(X,Y)/Var(X) • **a** = intercept = $\bar{y} - b * \bar{x}$ -- 4 marks **Best-Fit Line Determination:** • Compute b and a using formulas. • Draw line that minimizes total squared vertical distances from all data points to the line. -4M	4	7	Simple linear Regression and Correlation	Moderate	8.00	2
178	**Question:** What are the properties of a probability mass function (PMF)? **Solution and Scheme:** A PMF defines the probability distribution of a discrete random variable. Its properties are: 1. **Non-negativity:** P(X = x) ≥ 0 for all x 2. **Normalization:** ΣP(X = x) = 1 over all possible x 3. **Probability assignment:** P(X = x) gives the probability that the variable takes the value x	2	4	Discrete Probability distribution	Moderate	2.00	2
179	**Question:** Define and interpret a confidence interval with minimum 2 examples **Solution and Scheme:**	5	8	Statistical Inference for a single sample, Interval Estimation	Easy	8.00	2

	A **confidence interval** is a range of values, derived from sample statistics, that is likely to contain the population parameter. - 2 marks **Interpretation:** A 95% confidence interval means we are 95% confident that the true population mean lies within the interval. --- 2m 2 Examples -- 4 marks						
180	**Question:** What are stem and leaf diagrams? Illustrate with examples and mention its applications. **Solution and Scheme:** A **stem and leaf diagram** is a method of displaying quantitative data where each number is split into a “stem” (leading digit/s) and a “leaf” (last digit). -2M Example - 4marks Applications - 2 M	1	1	Data types and Charts	Easy	8.00	2
181	**Question:** State bayes theorem. **Solution and Scheme:** Bayes' theorem describes the probability of occurrence of an event related to any condition. It is also considered for the case of conditional probability. Bayes theorem is also known as the formula for the probability of “causes”. 2 marks	3	6	Sampling Distribution	Easy	2.00	1
182	**Question:** Mention any two characteristics of Binomial distribution. **Solution and Scheme:** Characteristics of a Binomial distribution: i. The number of observations n is fixed. ii. Each observation is independent. iii. Each observation represents one of two outcomes ("success" or "failure"). iv. The probability of "success" p is the same for each outcome. 4 points - 2 marks	2	4	Discrete Probability distribution	Moderate	2.00	2
183	**Question:** Define correlation coefficient **Solution and Scheme:** The correlation coefficient (r) is a statistic that tells you the strength and direction of that relationship. It is expressed as a positive or negative number between -1 and 1. The value of the number indicates the strength of the relationship: r = 0 means there is no correlation.	4	7	Simple linear Regression and Correlation	Easy	2.00	1
184	**Question:** Differentiate between Type I and Type II error **Solution and Scheme:** Rejection of the hypothesis when it should be accepted is known as Type I error. Acceptance of a hypothesis when it should be rejected is known as Type II error.	5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	2.00	4

185

Question:

A sample is collected to test if the population variance is more than 5. The hypothesis statements are _____ and _____

Solution and Scheme:

H_0: population variance =5

H_1: population variance >5

right answer - 2M

5	8	Statistical Inference for a single sample, Interval Estimation	Moderate	2.00	2

186

Question:

What do you meant by level of significance?

Solution and Scheme:

In testing a given hypothesis, the maximum probability with which we could be willing to risk is called level of significance of the test.

5	8	Statistical Inference for a single sample, Interval Estimation	Easy	2.00	1

187

Question:

Data have been obtained on the lives of batteries of a particular type in an industrial

application. Table 4.1 shows the lives of 36 batteries recorded to the nearest tenth of a year.

4.10	5.2	2.8	4.9	5.6	4	4.1	4.3	5.4
4.5	6.1	3.7	2.3	4.5	4.9	5.6	4.3	3.9
3.2	5	4.8	3.7	4.6	5.5	1.8	5.1	4.2
6.3	3.3	5.8	4.4	4.8	3	4.3	4.7	5.1

Construct a stem-and-leaf plot for the lives of the batteries.

Solution and Scheme:

Stem-and-Leaf Display:

Stem	Leaf	Frequency	Cum Frequency
1	8	1	1
2	8 3	2	3
3	7 9 2 7 3 3 0	6	9
4	1 9 0 1 3 5 5 9 3 8 6 2 4 8 3 7	16	(16)
5	2 6 4 6 0 5 1 8 1	9	11
6	1 3	2	2

1	1	Data types and Charts	Moderate	8.00	3

Question:

	0		Easy	1.00	0